

Analyzing heterogeneity in the effects of behavioral climate interventions on individual beliefs across sociodemographics, attitudes, and norms

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Abstract

Addressing climate change requires global shifts in individual beliefs at scale. While several behavioral interventions have been developed to strengthen climate change beliefs (e.g., by framing climate change as a psychologically proximal risk), the effectiveness of such interventions differs across individuals. Yet, the extent and drivers of such heterogeneity in intervention effectiveness remain largely unclear. Here, we aim to understand individual-level differences in the effectiveness of 11 expert-crowdsourced interventions on climate change beliefs across two randomized controlled trials (RCTs) (RCT 1: $N = 59,508$ individuals from 63 countries; RCT 2: $N = 1,610$). First, our analysis reveals large heterogeneity in the effectiveness: even the most effective intervention in terms of average treatment effect fails to motivate belief change for 20% of individuals, while interventions regarded as “ineffective” still positively influence 47%. Second, we identify key drivers across sociodemographics, attitudes, and norms, thereby showing that interventions effective in the Global North often backfire in the Global South, where different socioeconomic contexts shape responses. Third, we performed an online field study ($N = 772,777$) to externally validate the findings, showing that personalizing interventions increases effectiveness by more than 30%. Our findings emphasize the importance of personalization in behavioral climate interventions to maximize the effectiveness of behavioral strategies and foster widespread support of climate action.

1 Main

Climate change poses a global threat to society. Global temperatures in 2024 rose to approximately 1.5°C above pre-industrial levels, marking the hottest year on record [1]. Despite overwhelming scientific consensus that climate change is caused by human activity [2–4], public opinion remains largely misaligned with scientific evidence: 16% of Americans do not believe in climate change, and 39% do not support that citizens should take greater actions [5]. This disconnect has substantial negative consequences by weakening the support for climate policies and reducing motivation for individual and collective climate action [6, 7].

To prompt belief in climate change and climate action, one promising strategy that is scalable and cost-effective is through behavioral interventions [8–10]. These interventions aim to strengthen belief, for example, by framing climate change as a psychologically proximal risk [11], emphasizing future self-continuity [12], listing examples of successful collective action [13, 14], or evoking ingroup-loyalty and authority moral foundations [15]. The average effect of such interventions is well studied [16–19]. For example, a recent study compared 11 behavioral interventions aimed at targeting climate change beliefs and found that some interventions are more effective than others, yet the overall average effects were small [18, 20]. This pattern is consistent with recent meta-analyses across a wide range of climate mitigation actions, which similarly report modest average effects [16, 17]. However, by focusing only on average effects, one may overlook large variability in the effectiveness of behavioral interventions [21], such as that some interventions deemed ineffective are still highly effective for certain individuals or subgroups. Here, we thus aim to understand the heterogeneity in the effectiveness of behavioral climate interventions across individuals. Interestingly, we later show that some climate interventions generally considered ineffective are actually successful in promoting the belief for more than 40% of the study population.

Understanding how different individuals respond to behavioral climate interventions is crucial for maximizing their effectiveness [21]. In fact, focusing only on average effects ignores that indi-

viduals may respond differently to behavioral climate interventions and thus overlooks meaningful heterogeneity in their effectiveness, as positive effects for some individuals and negative effects for others may cancel out on average [21], thus creating a false impression that an intervention is ineffective. In addition, relying upon average effects risks reinforcing inequalities by prioritizing outcomes for majority groups, while minority groups, such as specific age cohorts or ethnic minorities, may respond differently and thus may not benefit [21]. To better target behavioral interventions, it has therefore been argued that research should move beyond average effects to identify which interventions work best for whom [21,22]. However, the heterogeneity in the effectiveness of behavioral interventions has been explored so far in the context of education [23], health [24], and social policies [25], but not for behavioral climate interventions.

The effectiveness of interventions is likely to vary across individuals due to individual differences [21, 26]. Human behavior is inherently heterogeneous and shaped by factors such as demographics, psychological traits, attitudes, and norms [9,27,28]. Given that there is variation in baseline climate beliefs across these factors [9,28–32], it is likely that interventions shift beliefs differently across groups. Recent work suggests that the effectiveness of climate interventions varies across political ideology [33] and cultural individualism [34]. However, it remains unclear how the effectiveness of climate interventions varies across a broader and more comprehensive range of individual characteristics, including sociodemographics, attitudes, and norms.

In this paper, we hypothesize that the effectiveness of climate interventions is heterogeneous and can be explained by individual characteristics (see Supplement C for a detailed theoretical motivation). Specifically, we focus on three dimensions: sociodemographics, psychological attitudes, and norms. First, sociodemographic factors such as age, gender, and education are known to shape baseline climate beliefs and risk perceptions [9,29,30], and different groups may perceive or react to climate interventions differently due to differential vulnerability, political affiliation, or historical exposure to environmental risks [35]. Second, psychological attitudes, such as environmental identity and trust in climate science, are key predictors of climate-related behaviors, specifically, by

shaping both motivation for action and responsiveness to climate messaging [30, 35]; individuals with strong pre-existing concerns or distress may be more receptive to interventions framed around collective action or emotional engagement. Third, perceived social norms strongly influence pro-environmental behavior [36, 37] and could amplify or dampen the effectiveness of norm-based interventions. By identifying how these dimensions shape intervention responses, we aim to move beyond average treatment effects (ATE) and toward a more differentiated understanding of which interventions work for whom by analyzing the heterogeneous treatment effects (HTE).

In this paper, we analyze the heterogeneity in the effectiveness of 11 expert-crowdsourced behavioral interventions designed to promote climate change beliefs (see Fig. 1). Our analysis comprises three major steps. First, we examine 11 expert-crowdsourced interventions from [18, 20] to quantify heterogeneity in their effectiveness across individuals. For this, we reanalyze the data from a large-scale, global randomized controlled trial (RCT) across 63 countries (RCT 1: $N = 59,508$). Here, we use causal machine learning [38, 39] to estimate heterogeneous treatment effects. We show empirically that substantial variation in treatment effectiveness is overlooked when analyzing average effects. That is, climate interventions with no or only small average effects can still be highly effective for certain subgroups. Second, we identify individual-level drivers of this effect heterogeneity. For this, we first reanalyze data from RCT 1 to identify individual-level drivers across sociodemographic variables (e.g., age and gender), and we then conduct an additional experiment (RCT 2: $N = 1,610$) to understand the underlying mechanisms related to attitudes (e.g., trust in climate science) and norms (e.g., perceived scientific consensus). Third, we develop a personalized targeting strategy based on the above analysis and externally validate it using a preregistered online field study on *Google Ads* ($N = 772,777$), where we measure the engagement with climate change content. This approach allows us to quantify the benefit of personalizing climate interventions to individuals.

Altogether, we find consistent evidence that traditional “one-size-fits-all” strategies for deploying behavioral climate interventions are ineffective and may even backfire. Interestingly, climate

96 interventions generally considered ineffective are actually successful in promoting climate change
97 belief for more than 40% of the study population. This challenges previous research, which pri-
98 marily evaluates interventions based on the ATE and thus ignores the differential effectiveness of
99 such interventions in subgroups such as females, older adults, and countries in the Global South.
100 To maximize effectiveness in promoting support for climate beliefs and action, a personalized
101 approach is necessary, which accounts for the sociodemographic, attitudinal, and normative char-
102 acteristics of the recipient population.

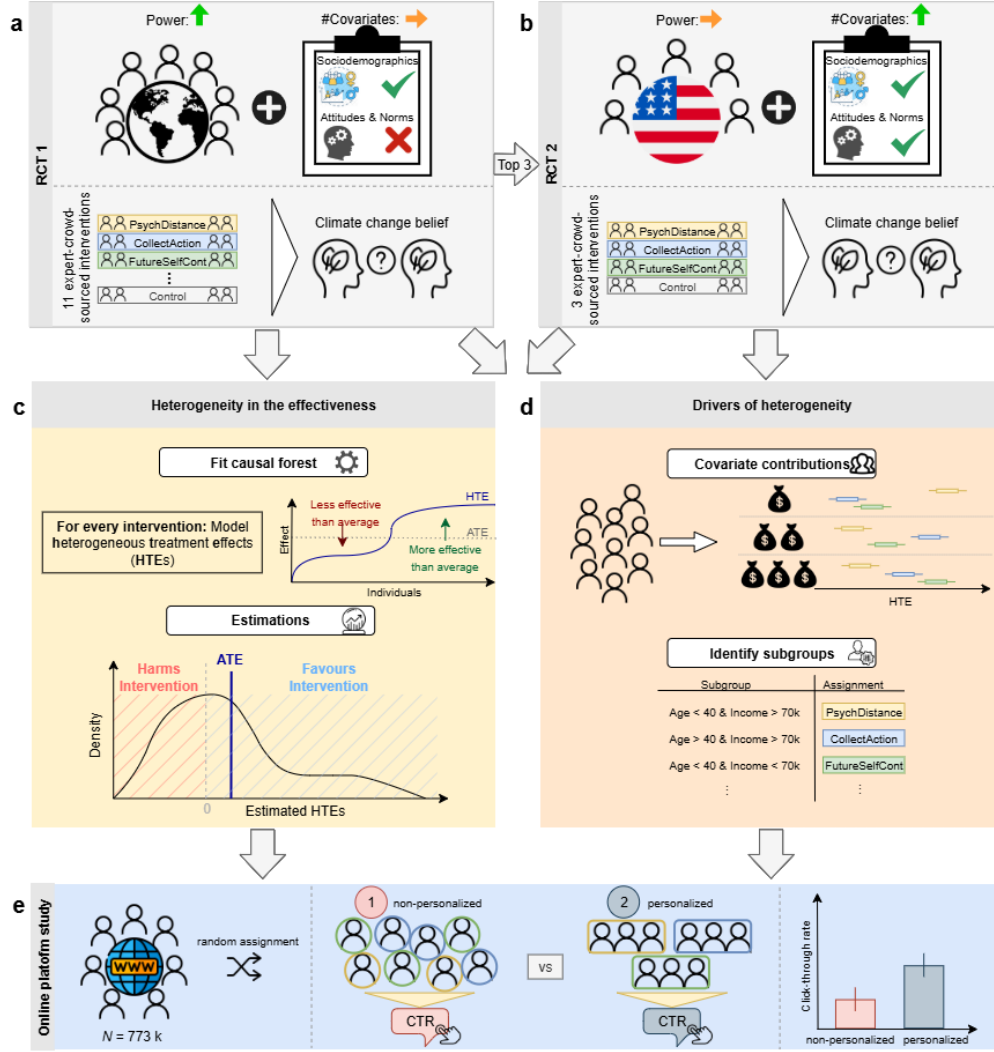


Figure 1: Workflow for analyzing heterogeneity in the effectiveness of behavioral climate interventions. We quantify heterogeneity in the effectiveness of 11 expert-crowdsourced interventions designed to promote climate change beliefs. Climate change belief is chosen as the primary outcome, as it predicts downstream behaviors such as policy support [40–43]. **a**, The strengths of RCT 1 ($N = 59,508$) are the high statistical power and the broad geographic coverage across 63 countries. **b**, RCT 2 ($N = 1,610$) gathers additional information about attitudes and norms to understand psychosocial mechanisms. **c**, To analyze heterogeneity in effectiveness, we use a causal machine learning model (i.e., causal forest [38, 39]) to estimate heterogeneous treatment effects (HTEs). This allows us to contrast the average treatment effect (ATE) of climate interventions against how the effectiveness varies across individuals. **d**, We leverage insights from our causal forest to analyze drivers of effect heterogeneity across individual-level sociodemographics, country-level sociodemographics, attitudes, and norms. We then identify subgroups that respond differently to behavioral climate interventions. **e**, Using an online platform study, we test our personalized targeting strategy ($N = 772,777$). We test our personalized targeting strategy on *Google Ads* to measure user engagement with climate change content. This allows us to quantify the relative benefit of personalization compared to a standard, non-targeted strategy.

2 Results

To estimate the HTEs for each intervention, we employ a causal forest [38, 39], which is a state-of-the-art causal machine learning method for estimating HTEs by modeling complex, high-dimensional interactions between variables to capture heterogeneity in the treatment effects [38, 39]. The HTEs then capture the estimated difference in climate change belief for each specific intervention as compared to receiving no intervention (i.e., being in the control condition), conditional on individual-level variables. This approach allows us to identify which intervention is estimated to be most effective for each individual based on their characteristics (see the Methods section for more details). To interpret the heterogeneity, we later identify key drivers by comparing the characteristics of the individuals across the estimated HTEs and use these insights to define subgroups for which specific interventions are estimated to work best. We use these subgroups to inform our personalized targeting strategy, which we then externally validate based on a preregistered online study on *Google Ads*.

2.1 Heterogeneity in the effectiveness of behavioral climate interventions

To understand how the effectiveness of behavioral climate interventions varies across individuals, we estimate heterogeneous treatment effects (HTEs) using data from a large-scale randomized controlled trial (RCT 1). This RCT involved 59,508 participants from 63 countries who were randomly assigned to receive one of 11 expert-crowdsourced behavioral interventions or a control condition [18, 20]. The interventions were designed to increase belief in climate change and include, for example, *framing climate change as a psychologically proximal risk* (PSYCHDISTANCE), *emphasizing future self-continuity* (FUTURESELFCONT), or *listing examples of successful collective action* (COLLECTACTION) (see Supplementary Table S1 for a full list). Our outcome of interest is a composite measure of climate change belief, averaged across four Likert-scale items, which was scaled from 0 to 100. Thus, an HTE of 10 indicates that the intervention is estimated to increase

climate change belief by 10 points on this scale, given an individual’s characteristics.

The HTEs reveal substantial heterogeneity in how individuals respond to the interventions (see Fig. 2). Interestingly, for all interventions, the distributions of the HTEs are wide and span both positive and negative effects, indicating that, even when ATEs are small, many individuals may still benefit substantially. For example, *framing climate change as a psychologically proximal risk* (PSYCHDISTANCE) shows the highest ATE overall and succeeds in motivating positive change in belief for 80% of individuals but still backfires for 20% of individuals. Similarly, *emphasizing future self-continuity* (FUTURESELFCONT) and *listing examples of successful collective action* (COLLECTACTION) lead to positive HTEs for 76% and 62% of participants, respectively. Interestingly, *work-together normative appeals* (IDENTITYSOCIALNORMS) was often regarded as ineffective in prior literature [18,33] due to the negative ATE (i.e., -0.65), but our analysis reveals that this intervention still benefits 47% of the study population. Others, such as *writing a letter to future generation* (LETTER2FUTURE) and *work-together normative appeals* (IDENTITYSOCIALNORMS), show bimodal patterns, helping some subgroups substantially while showing limited or even negative effects for others.

These findings highlight the limitations of relying solely on ATEs to evaluate behavioral interventions. As we have shown above, a modest ATE may conceal the fact that a substantial share of individuals may respond positively. For example, although NEGATIVEEMOTIONS and (BINDINGMORAL) show near-zero ATEs, they still benefit 47% and 45% of participants, respectively. Conversely, even the most effective interventions, such as PSYCHDISTANCE, may be ineffective for specific subgroups (e.g., 20% with negative HTEs). Taken together, these results highlight the value of analyzing HTEs alongside ATEs, as doing so reveals that many behavioral interventions help far more people than aggregate metrics suggest. These patterns motivate a more nuanced question in the following section: *for whom do interventions work best?* This aligns with growing calls in behavioral science to better understand individual-level heterogeneity [21,44] when evaluating the effect of behavioral interventions.

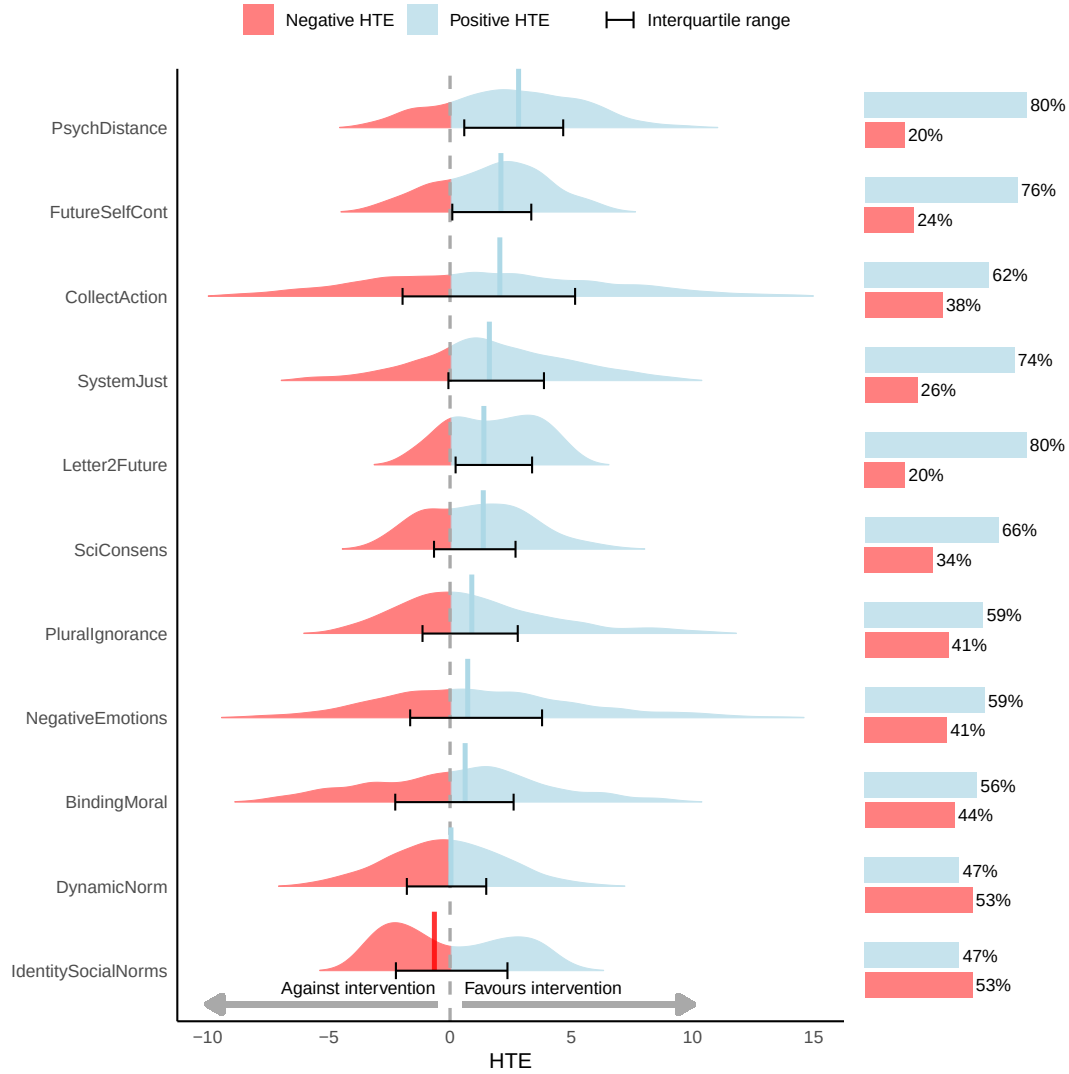


Figure 2: Estimated HTEs across interventions and individuals. The figure shows the distribution of estimated heterogeneous treatment effects (HTEs) for each of the 11 behavioral climate interventions, based on experimental data from $N = 59,508$ participants across 63 countries. Each HTE represents the estimated difference in climate change belief compared to a control condition, conditional on individual variables. The bar charts indicate the proportion of individuals for whom an intervention is estimated to have a positive (blue) or negative (red) effect. For example, *framing climate change as a psychologically proximal risk* (PSYCHDISTANCE) benefits an estimated 80% of individuals, while 20% show no or negative change. These distributions highlight that intervention effectiveness varies widely across individuals. For ease of interpretation, we indicate the interquartile range (horizontal whiskers) and the average treatment effect (ATE; vertical line) estimated using a doubly robust augmented inverse propensity weighting (AIPW) method [45,46]. For a full description of interventions, see Supplementary Table S1.

Differences across Global North versus Global South. We also observe substantial differences in the effectiveness of the interventions when comparing individuals from the Global North vs. the Global South (see Fig. 3). For each intervention, the corresponding distributions of the HTEs are significantly different from each other (Kolmogorov–Smirnov test: $D > 0.15$, $p < 0.001$), with clear shifts in the median HTE. For example, PSYCHDISTANCE shows a high median HTE (3.15) for individuals in the Global North relative to the median for the Global South (0.38). In contrast, the least effective intervention on average, namely IDENTITYSOCIALNORMS, has a strong positive median HTE (2.56) for individuals from the Global South compared to Global North (−1.69). These findings highlight that intervention strategies that may work well in the Global North cannot simply be transferred to the Global South. Additional analyses across country-level characteristics (Supplementary Figures 3 and S4 to S7) further substantiate the observed heterogeneity and identify key drivers associated with HTEs.

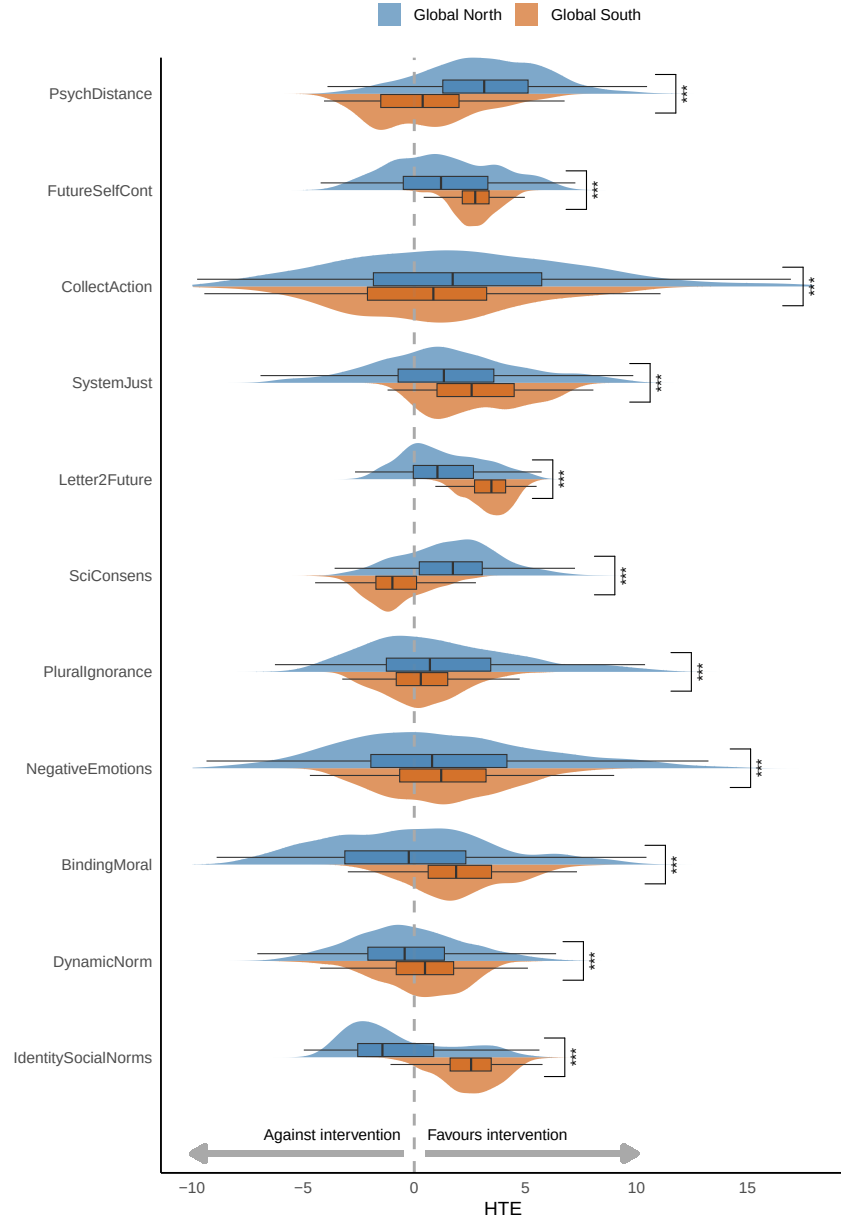


Figure 3: Estimated HTEs across interventions and individuals from the Global North vs. the Global South. The figure shows the distribution of estimated heterogeneous treatment effects (HTEs) for each of the eleven behavioral climate interventions for countries from the Global South vs. Global North, based on experimental data from RCT 1 ($N = 59,508$ participants across 63 countries). Each HTE represents the estimated difference in climate change belief compared to a control condition, conditional on individual variables. We compare the distributions of HTEs between the Global South versus the Global North within each intervention using a two-sample Kolmogorov-Smirnov test, with significance thresholds indicated as: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. Box plots show the different quartiles. Overall, we observe that the distributions differ significantly for all interventions, indicating that intervention strategies that may be effective in the Global North cannot be simply transferred to the Global South. For a full description of interventions, see Supplementary Table S1.

Robustness Checks. To assess the robustness of our model estimates, we conducted several out-of-sample validations of the estimated HTEs. Following best practices in causal machine learning, we applied the effect scores method [47–49] to evaluate whether the ordering of estimated HTEs aligned with observed treatment effects across subgroups. Specifically, we computed HTEs on a held-out test set, sorted individuals into quartiles based on their estimated HTEs, and then estimated the ATE within each quartile using augmented inverse propensity weighting (AIPW) [45, 46]. Across all interventions, ATEs increased monotonically from the lowest to the highest quartile (see Supplementary Figure S2), which demonstrates that the causal forest model can successfully capture heterogeneity in the treatment effects and further provides strong evidence that the effect scores are well-calibrated. We also compared the HTE estimates of our model to those obtained from another causal machine learning approach, the T-learner [50], where we find that the HTE estimates are highly correlated (Pearson’s correlation coefficient: $r > 0.4$, $p < 0.001$ for all interventions; see Supplementary Figure S3), which further confirms the robustness of our model.

2.2 Drivers of heterogeneity

We now analyze the drivers of heterogeneity in the effectiveness of behavioral climate interventions. For this, we first reanalyze data from RCT 1 to identify sociodemographic drivers, and second, we conduct an additional experiment (RCT 2: $N = 1,610$) to identify attitudinal and normative drivers. In the former, we analyze a set of 10 potential drivers with sociodemographics captured at the individual (e.g., gender and age) or country level (e.g., the Gini index). See Supplementary Table S2 for a full list and description. In the latter, we analyze the underlying drivers across 10 additional variables related to attitudes (e.g., trust in climate science) and norms (e.g., perceived climate change belief). The full list of the variables related to attitudes and norms is provided in Supplementary Table S3. Our selection of potential drivers of heterogeneity is informed by prior literature [18, 30, 33, 51, 52] and the goal of capturing a broad range of variables

that possibly shape how individuals respond to behavioral climate interventions.

We use a two-stage approach to identify key drivers of heterogeneity. First, we examine how individual characteristics are distributed across people with low, medium, and high estimated HTEs. This grouping is based on sorting individuals into tertiles of their estimated HTEs. By comparing variable values across these groups, we assess whether certain characteristics are systematically associated with higher intervention effectiveness. For statistical testing, we use the Kruskal-Wallis test [53], a non-parametric method that assesses whether variables of the individuals across the different tertiles likely come from the same distribution. A significant result indicates systematic heterogeneity in the variables across tertiles, implying that the characteristics of individuals are different for treatment responders vs. non-responders. Second, we identified which behavioral intervention works best for each subgroup by computing the share of individuals for whom a specific intervention is estimated to be optimal, that is, estimated to have the highest HTE. This allows us to determine the most effective intervention strategy for each subgroup. For clarity, we report results for the three most promising interventions based on the ATEs in RCT 1, namely, (1) PSYCHDISTANCE, (2) FUTURESELFCONT, and (3) COLLECTACTION. The full results for all other interventions are available in Supplementary Figure S8.

Sociodemographic drivers

Across all three interventions analyzed, 26 out of 30 variable-intervention combinations show statistically significant differences at conventional significance levels across tertiles sorted by the estimated HTEs (see Fig. 4). Notably, the three behavioral climate interventions are consistently more effective for individuals with lower socioeconomic status, measured by education and income. In addition, individuals in the upper tertile of estimated HTEs tend to have a higher proportion of females than average. In contrast, variables such as age exhibit more intervention-specific patterns, which suggests that sociodemographic drivers of heterogeneity interact with the content of the behavioral climate intervention. For example, individuals in the lower tertile of estimated HTEs for

FUTURESELFCONT are older on average ($H = 43.42$; $p < 0.001$). In contrast, the lower tertile for COLLECTACTION consists of younger individuals.

Individuals from countries with higher CO₂ emissions per capita tend to fall into higher HTE tertiles across all interventions (all $H > 266$; $p < 0.001$), possibly reflecting greater contextual salience of climate risk. Similarly, most interventions tend to be more effective for individuals from countries with higher levels of climate change risk and income inequality (measured by the Gini index [54]). Surprisingly, for PSYCHDISTANCE, individuals from higher-risk and higher-inequality contexts fall into the lower HTE tertile, implying that PSYCHDISTANCE is less effective or may even backfire for them. One possible explanation is that, in countries already facing high climate risk or social inequality, framing climate change as a psychologically proximal risk may not introduce new information for belief change, while in lower-risk contexts, it may be more novel. Interestingly, these country-level patterns map closely onto the Global North–Global South divide and point to the nuanced role of country-level sociodemographics in driving intervention effectiveness (see Supplementary Figures S4 to S7 for a detailed analysis of country-level heterogeneity). Altogether, our findings motivate the case for tailored strategies when deploying behavioral climate interventions.

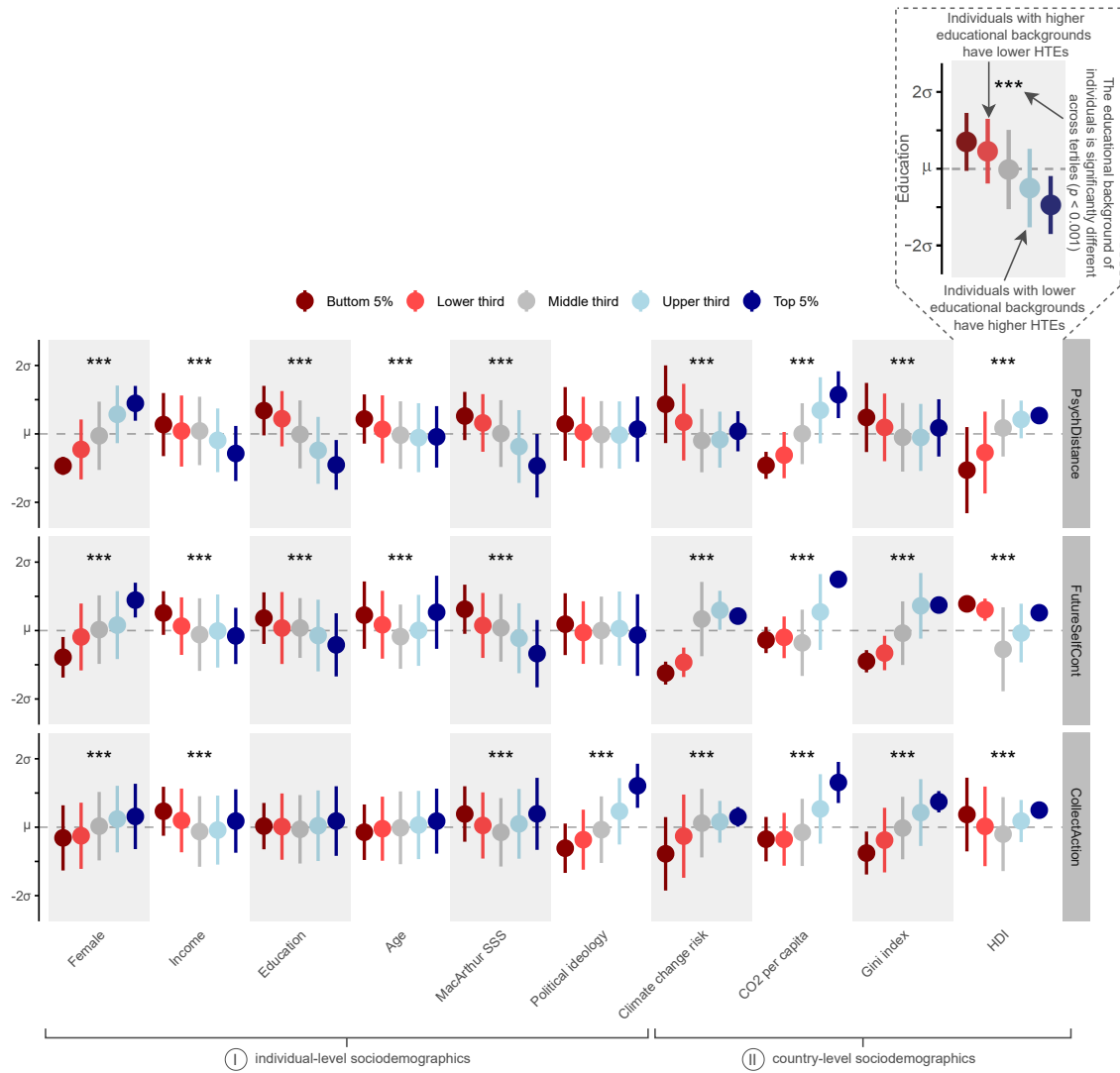


Figure 4: Sociodemographic drivers of heterogeneity in the effectiveness of behavioral climate interventions. Here, we assess whether certain characteristics are systematically associated with higher intervention effectiveness. The plot shows how the sociodemographic variables, including the MacArthur scale of subjective social status (MacArthur SSS) and the human development index (HDI), are distributed across individuals with different estimated effects of behavioral climate interventions. Here, we stratify the individuals into tertiles by the estimated HTEs for the three selected interventions: PSYCHDISTANCE, FUTURESELFCONT, and COLLECTACTION. In addition to the tertiles, we also show the bottom and upper 5% tails of the estimated HTEs. Each point represents the standardized mean of a variable within a group, and the error bars show the standard error. Variables are standardized (mean = 0, SD = 1) to allow for comparability across scales. Hence, a point at $+1\sigma$ means that the average value of the variable in the specific tertile is one standard deviation higher than the overall population mean. For example, when individuals with lower levels of educational background fall into higher HTE tertiles, this implies that the corresponding intervention is more effective for less-educated individuals. Differences across tertiles are tested using the non-parametric Kruskal-Wallis test [53], with significance thresholds indicated as: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. Results for the other interventions are in Supplementary Figure S8.

We next examine which behavioral climate intervention is estimated to be most effective across different sociodemographic groups (see Fig. 5). For each subgroup, we calculate the proportion of individuals for whom a given intervention yields the highest estimated HTE on climate change belief, based on estimations of our causal forest model on a held-out test set. The optimal behavioral intervention varies systematically across individual-level sociodemographic subgroups, thereby revealing large differences in which subgroups respond best to which intervention (see Fig. 5). Interestingly, 84.1% of females are estimated to respond best to PSYCHDISTANCE and COLLECTACTION, whereas the distribution across interventions is more balanced for males. This suggests that emotionally grounded and community-oriented framings may resonate more strongly with women. Also, individuals with lower levels of education are more likely to respond best to PSYCHDISTANCE (63.3%), whereas those with higher levels of education are more responsive to COLLECTACTION (49.1%). Here, a possible reason might be that more highly educated individuals are more persuaded by efficacy-based framing that emphasizes successful collective efforts. Finally, among conservatives, COLLECTACTION is often the optimal intervention (58.6%), while only 19.3% of liberals respond best to it, suggesting that the effectiveness of climate interventions varies across ideological lines.

We observe substantial variation in the optimal interventions across country-level sociodemographic subgroups. Notably, PSYCHDISTANCE is often optimal in countries with low climate change risk (76.5%), but much less so in high-risk countries (11.5%). This pattern may reflect an awareness effect: in low-risk countries, framing climate change as a proximal threat may prompt a positive change in belief, whereas in high-risk countries, such climate threats are already part of daily experience and may no longer lead to substantial shifts. This is in line with previous findings showing that initial climate change beliefs are highest in countries with high climate change risk, such as Nigeria, Peru, and Mexico [18]. Similarly, PSYCHDISTANCE is more often optimal in countries with a low Gini index (67.4%) and high human development index (HDI) (66.4%), potentially because individuals in more equal and developed societies may be more psychologi-

257 cally responsive to narrative-based framings that link climate events to personal impact. In con-
258 trast, FUTURESELFCONT may resonate more strongly in low-HDI countries by offering a sense
259 of long-term hope that may otherwise be underrepresented in the climate discourse (i.e., FUTURE-
260 SELFCONT is often optimal in low-HDI countries (55.8%)). Altogether, these findings emphasize
261 that the effectiveness of behavioral climate interventions not only varies across individual-level so-
262 ciodemographics but also across the broader social and developmental context in which individuals
263 are embedded.

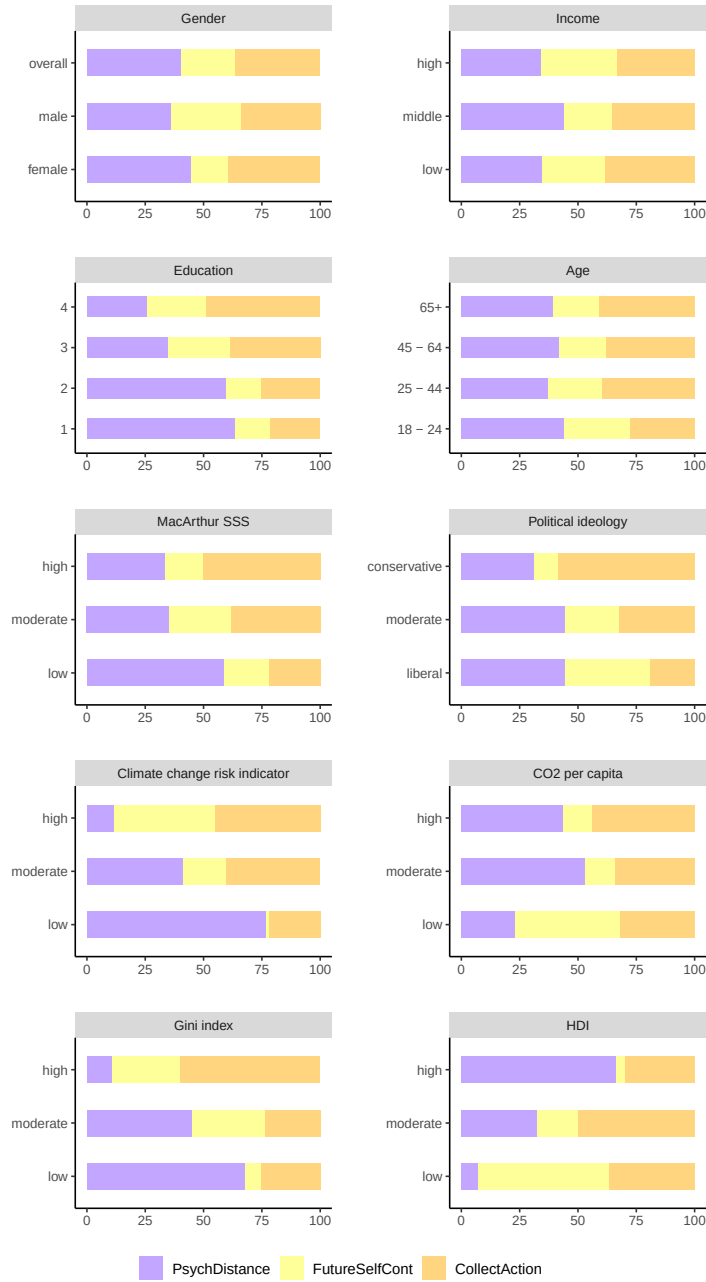


Figure 5: Optimal interventions by sociodemographic subgroups. The figure shows the proportions of how often each intervention is optimal (x-axis, from 0 to 100%) and therefore has the largest HTE within that subgroup as determined by the causal forest model. In other words, the optimal intervention is defined as the one with the highest HTE on climate change belief. Education levels are encoded using a discrete categorization (i.e., 1: up to grade school, 2: up to high school, 3: college or undergraduate degree/certificate training, 4: doctorate, medical, or other advanced degrees). We categorize the other continuous variables into three equally sized subgroups (e.g., high, middle, and low income). We denote the MacArthur scale of subjective social status as MacArthur SSS and the human development index as HDI. For a full description of the sociodemographics, see Supplementary Table S2.

Robustness Checks. To assess the robustness of our findings, we repeated the analysis using SHAP values [55], a method that explains the contributions of each sociodemographic variable to the HTE estimates of our causal forest. We find consistent patterns regarding the drivers of heterogeneity (see Supplementary Figure S10).

Attitudinal and normative drivers

We now examine whether individuals with low versus high estimated intervention effects differ systematically in their attitudes and norms, by analyzing how these characteristics are distributed across tertiles of the estimated HTEs (see Fig. 6). For example, the PSYCHDISTANCE intervention is more effective for individuals with lower values across attitudes and norms, such as lower environmental identity ($H = 96.55$; $p < 0.001$) and lower internal environmental motivation ($H = 98.97$; $p < 0.001$). This pattern may be explained by the psychological mechanism of PSYCHDISTANCE, which bridges psychological distance more effectively when individuals have not yet developed strong environmental attitudes and norms, rather than competing with existing ones. In contrast, the COLLECTACTION intervention shows largely opposite effects across attitudes and norms. COLLECTACTION is more effective for individuals with higher external environmental motivation ($H = 11.61$; $p < 0.01$), greater pluralistic ignorance ($H = 20.14$; $p < 0.001$), and higher trust in government ($H = 32.12$; $p < 0.001$). For FUTURESELFCONT, individuals in the top tertile of estimated HTEs have, on average, lower scores of the new ecological paradigm ($H = 40.33$; $p < 0.001$), suggesting that FUTURESELFCONT may be most effective when individuals have not internalized comprehensive environmental worldviews.

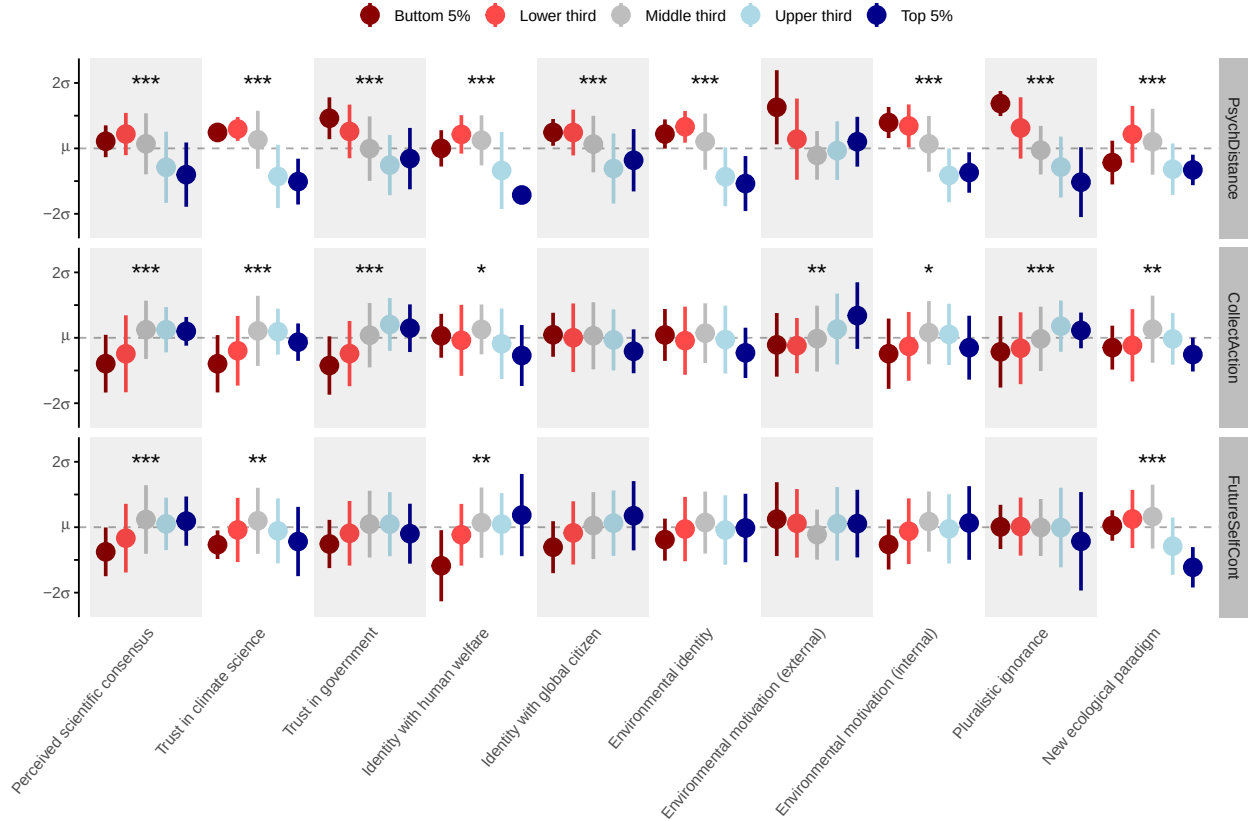


Figure 6: **Attitudinal and normative drivers of heterogeneity in the effectiveness of behavioral climate interventions.** Here, we assess whether there is a systematically higher intervention effectiveness for certain attitudes or norms. The plot shows how attitudes and norms, including the environmental identity and the new ecological paradigm, are distributed across individuals. Here, we stratify the individuals into tertiles by the estimated HTEs for the three selected interventions: PSYCHDISTANCE, FUTURESELFCONT, and COLLECTION. In addition to the tertiles, we also show the bottom and upper 5% tails of the estimated HTEs. Each point represents the standardized mean of a variable within a group, and the error bars show the standard error. Variables are standardized (mean = 0, SD = 1) to allow for comparability across scales. A point at $+1\sigma$ means that the average value of the variable in the specific tertile is one standard deviation higher than the overall population mean. Differences across tertiles are tested using the non-parametric Kruskal-Wallis test [53], with significance thresholds indicated as: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

The optimal behavioral intervention varies across groups of attitudes and norms (see Fig. 7). Generally, subpopulations with lower scores on attitudes and norms are often estimated to respond best to PSYCHDISTANCE. For example, 86.4% of individuals with low levels of perceived scien-

287 tific consensus respond best to PSYCHDISTANCE, suggesting that psychological distance framing
288 may be particularly effective when baseline environmental beliefs are less established. In con-
289 trast, COLLECTACTION demonstrates higher percentages of performing best for individuals with
290 higher attitudinal and normative levels. For instance, COLLECTACTION is optimal for 40.0%
291 of individuals with high levels of external environmental motivation. This pattern suggests that
292 collective action messaging leverages external motivational structures to drive belief change. Sim-
293 ilarly, the highest proportion of individuals responding best to FUTURESELFCONT emerges within
294 subgroups exhibiting the highest attitudinal and normative levels. Here, 35.8% of individuals with
295 high perceived scientific consensus respond best to FUTURESELFCONT, suggesting that future
296 self-continuity is more effective when individuals already possess strong foundational environ-
297 mental beliefs.

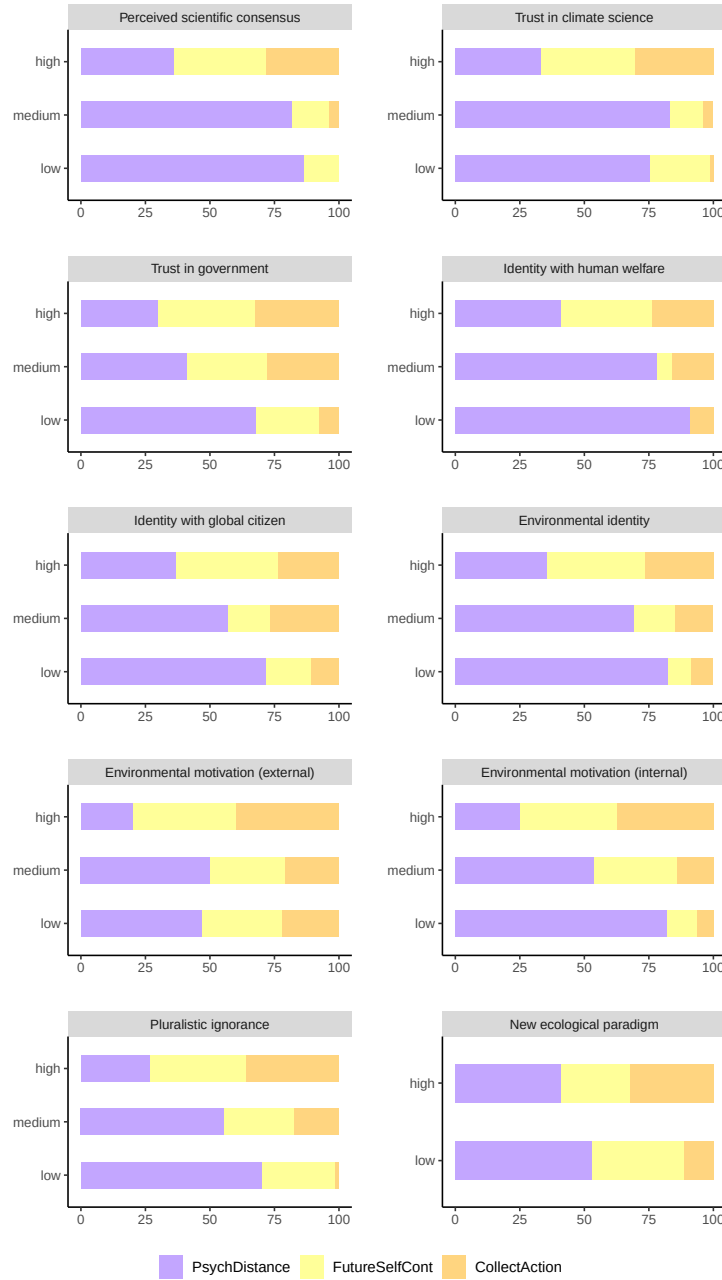


Figure 7: **Optimal intervention by attitudinal and normative subgroups.** The proportions of how often each intervention is optimal (x-axis, from 0 to 100%) and therefore has the largest HTE within that subgroup as determined by the causal forest model. In other words, the optimal intervention is defined as the one with the highest HTE on climate change belief. We categorize the continuous variables into three equally sized subgroups (e.g., high, middle, and low-level environmental identity). For a full description of the sociodemographics, see Supplementary Table S3.

Robustness Checks. We again repeated the analysis using SHAP values [55] to explain the contributions of each attitudinal and normative variable to the HTE estimates of our causal forest. We find consistent patterns regarding the drivers of heterogeneity (see Supplementary Figure S11).

2.3 External validation using an online platform study

To externally validate our findings for personalization strategies, we conduct a preregistered online platform study on *Google Ads* ($N = 772,777$), which allows us to assess the ecological validity of our previous analyses in a real-world setting. Here, we test whether subgroups identified based on our previous analyses can inform effective targeting strategies (personalized approaches) in real-world settings to increase engagement (measured through clicks) with climate change content at scale beyond one-size-fits-all strategies (non-personalized approaches). Specifically, we use our framework from above to assign optimal behavioral interventions per subgroup based on estimated HTEs for our personalized approach and compare the click-through rate (CTR) against a non-personalized approach that assigns behavioral interventions to individuals, independent of their characteristics.

To obtain a targeting strategy for our personalized approach, we re-estimate the causal forest model using the U.S. sample from RCT 1 and then identify the optimal intervention for the different subgroups defined by age and gender. The reason for this is that *Google Ads* allows targeting only by selected variables (i.e., gender and age). Our targeting strategy assigns the following interventions to each subgroup: females aged between 18 and 24 years receive PSYCHDISTANCE; females aged between 25 and 44 years receive COLLECTACTION; females aged between 45 and 64 years receive COLLECTACTION; females aged 65 years or older receive FUTURESELFCONT; males aged between 18 and 24 years receive PSYCHDISTANCE; males aged between 25 and 44 years receive COLLECTACTION; males aged between 45 and 64 years receive PSYCHDISTANCE; males aged equal or above 65 years receive COLLECTACTION. We report the detailed estimation results for the HTEs used to inform our personalized targeting strategy in Supplementary Fig-

ure S12. In contrast, the non-personalized approach randomly assigns a behavioral intervention to each individual, irrespective of the estimated HTE.

We use a logistic regression to estimate the effect of our personalized approach on the probability of users clicking on climate change content (see Methods). Overall, we find a significant effect of using our personalized approach over the non-personalized approach (see Fig. 8a). Specifically, using the personalized approach increases the odds of clicking by 31.35% (95% CI: [26.28%, 36.62%], $p < 0.001$). At the aggregate level, the CTR for the non-personalized approach is 1.15%, while the CTR for the personalized approach is 1.50%, which corresponds to a relative increase of 30.43%. In addition, we find that the personalized approach significantly increases the odds of clicking on climate change content for six out of the eight predefined sociodemographic subgroups (see Fig. 8b). Specifically, the effects are strongest for females aged 45-64 (+29.71%, $p < 0.001$) and males aged 65+ (+126.99%, $p < 0.001$). We find no significant difference for the subgroups of females aged 18-24 and males aged 45-64. Altogether, our findings show that personalizing behavioral climate interventions by matching individuals to the interventions with the largest estimated HTE is highly effective.

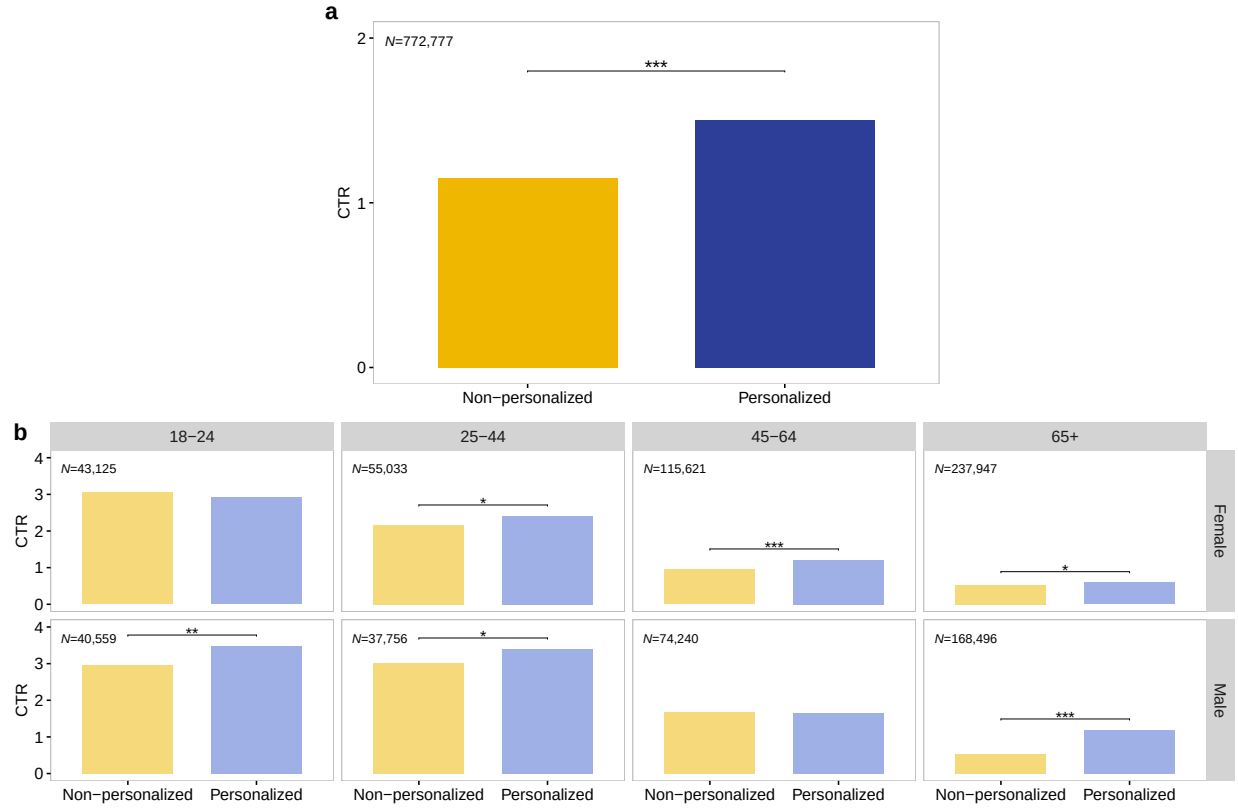


Figure 8: External validation of the personalized targeting strategy. Results based on our on-line platform study on *Google Ads* with $N = 772,777$ individuals, where N refers to the number of unique impressions (number of users who were presented with an ad). **a**, The bar chart represents the click-through rates (CTRs) for the non-personalized approach (in yellow) vs. the personalized approach (in blue) at the aggregated level. **b**, The bar charts represent CTRs for each sociodemographic subgroup. The columns (e.g., 18-24) refer to the age category, and rows refer to the gender category. We test for statistically significant differences using a logistic regression (detailed estimation results are in Supplementary Table S4). Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

3 Discussion

Our analysis of expert-designed behavioral climate interventions across two RCTs (RCT 1: $N = 59,508$ individuals from 63 countries; RCT 2: $N = 1,610$ from the U.S.) reveals substantial heterogeneity in their effectiveness. Oftentimes, behavioral climate interventions are considered ineffective in prior literature [18], while, interestingly, our analysis of HTEs demonstrates that even the least effective intervention in terms of ATE is actually successful in promoting the belief in climate change among more than 40% of the study population. While some interventions, such as PSYCHDISTANCE and FUTURESELFCONT, have higher ATEs than others, we also observe that they may backfire for large parts of the society. Importantly, the distributions of HTEs differ substantially across contexts, with generally wider spreads and shifted medians in the Global North compared to the Global South. While being the most effective intervention on average, PSYCHDISTANCE shows a near-zero median HTE in the Global South (0.38) compared to a markedly higher median in the Global North (3.15), illustrating how the same intervention can have very different effects across contexts. These results show that an analysis focused on the ATE can mask important individual-level differences in the response to climate interventions and highlights the value of estimating HTEs to uncover who benefits—and who may be adversely affected—by a given intervention.

Climate interventions were generally more effective among individuals with lower socioeconomic status, measured by education and income. Yet, the effectiveness varied considerably across sociodemographics, attitudes, and norms, with unique, intervention-specific patterns. For example, PSYCHDISTANCE was more effective among individuals with less education and females, while COLLECTACTION showed higher effectiveness for those with more education. Similarly, country-level factors also shaped effectiveness. Importantly, different from the other interventions, PSYCHDISTANCE was less effective in countries with greater climate change risks, greater inequality, and lower CO₂ emissions, which are prevalent conditions in the Global South. Here, it

is likely that proximal framing adds little new information as climate threats are already highly visible. These findings suggest that intervention strategies that may work well in the Global North cannot simply be transferred to the Global South. As a result, the optimal intervention varies greatly across subgroups, which highlights the need for personalizing interventions to individuals. For instance, PSYCHDISTANCE and COLLECTACTION together were optimal for nearly all females, while males showed more mixed patterns. In contrast, FUTURESELFCONT was often the most effective intervention in low-HDI countries, which are mostly located in the Global South. Attitudinal and normative variables such as environmental identity, trust in climate science, and perceived scientific consensus also meaningfully moderated effectiveness. Here, individuals with lower levels of attitudes and norms consistently responded better than average to PSYCHDISTANCE, whereas we mostly see a contrary pattern for COLLECTACTION. Together, these findings highlight the limitations of one-size-fits-all approaches and make a strong case for tailored climate communication strategies that account not only for sociodemographic factors such as age, gender, and income, but also for the regional and cultural differences between populations in the Global North and the Global South.

Our findings offer novel theoretical insights into the psychological mechanisms that may explain the heterogeneity in the effectiveness of behavioral climate interventions. The low effectiveness of the PSYCHDISTANCE intervention among individuals from high climate-risk countries is contrary to conventional expectations. Also, PSYCHDISTANCE is less effective across individuals who are psychologically engaged with climate change, demonstrated by a higher environmental identity or stronger internal environmental motivation. These findings can be explained by the Construal Level Theory [56], stating that people are more likely to already have high baseline beliefs of issues that are psychologically close rather than abstract or distant. Hence, in such settings, emphasizing psychological proximity offers little novelty and may convey information that these individuals may already implicitly understand or live with daily. The differential effectiveness of the COLLECTIVEACTION intervention appears to operate through mechanisms of collective effi-

cacy. Prior research shows that belief in collective agency [57] can foster motivation to engage in climate action [58]. For individuals embedded in strong social groups or with a strong sense of civic responsibility, collective framing can reinforce the belief that personal actions contribute meaningfully to shared outcomes. Finally, FUTURESELFCONT was less effective among older individuals, which is consistent with evidence that older adults have already formed their identity and are less responsive to messages that highlight long-term self-reflection [59]. Likewise, older individuals may also have a stronger temporal self-concept. Hence, in comparison, increasing the salience of the future self for younger individuals can bridge the psychological gap between present and future identity more effectively, thereby motivating long-term decisions like climate-friendly behavior [12].

Our findings contribute to and extend existing theoretical frameworks in several ways. First, they contribute to a growing body of research on heterogeneity in the effectiveness of behavioral interventions (e.g., [23–25]), which increasingly emphasizes the importance of moving beyond average effects. While prior work has examined HTEs in areas such as education [23], health [24], and social policies [25], similar, large-scale evidence for behavioral climate interventions is lacking. Here, we thus perform a novel study to examine heterogeneity across sociodemographic, attitudinal, and normative characteristics. Second, our results reinforce a broader narrative that understanding treatment effect heterogeneity is crucial for drawing accurate conclusions about behavioral interventions [21, 26]. Effects that appear small or even negative on average may be substantively large for particular subgroups with certain sociodemographic factors, attitudes, or norms (e.g., the intervention PSYCHDISTANCE is more effective for individuals with lower environmental identity and lower internal environmental motivation, while the effects are opposite for COLLECTIVEACTION). Finally, our findings connect to the discussion around personalization in behavioral science [60, 61] but offer novel evidence quantifying the benefit of personalized targeting strategies in a climate context, where interventions are matched to individuals based on estimated HTEs. In our online platform study, we find that personalizing climate change interventions can increase en-

gagement (as measured by CTR) by 30.43% compared to non-personalized approaches. Hence, for policy-makers, researchers, and practitioners, these results show the scientific and practical value of moving from generic to targeted behavioral interventions.

A key strength of our study lies in the comprehensive analytical framework for uncovering heterogeneity in the effectiveness of behavioral interventions, combined with rigorous empirical validation. First, our framework supports theory building by identifying effect modifiers—i.e., variables that explain for whom and under what conditions behavioral climate interventions are effective. As a result, our framework can also be used to develop personalized targeting strategies. Second, we use the causal forest as a flexible, state-of-the-art, nonparametric machine learning method [38, 39] to estimate individual-level HTEs across a broad, high-dimensional set of sociodemographic, attitudinal, and normative variables. This approach allows us to detect complex interaction patterns without relying on restrictive assumptions about the functional form, which is in contrast to linear interaction models, which typically depend on strong a priori assumptions about how interventions relate to outcomes. Third, unlike linear interaction models, the causal forest is further specifically designed to handle the bias–variance tradeoff, which makes causal forests especially useful in small-sample size settings with many covariates [62]. Similarly, due to the bias-variance tradeoff, the HTE estimates from the causal forest are designed to generalize well to new, unseen individuals, which is relevant when developing personalized targeting strategies. Fourth, we implement various internal validation strategies to ensure the robustness of our findings (e.g., such as those recommended in [47–49]). Fifth, we externally validate our findings by evaluating how well the estimated treatment effects generalize to a real-world online experiment with unseen individuals. Together, our framework offers a practical blueprint for estimating HTEs of behavioral interventions in other disciplines.

Our study is subject to several limitations, which present impactful paths for future research. First, we focused on climate change belief as our primary outcome because it is a known predictor of downstream behaviors [40–43]. We also analyze a related outcome, namely, climate

policy support (see Supplementary Figures S16 to S20). Future research could further explore other outcomes, including behavioral and real-world climate-relevant actions (e.g., adoption of climate tech [63]). Second, the options available for personalization on *Google Ads* are restricted to a limited set of variables (i.e., age and gender) due to privacy and legal reasons. As a result, other variables (such as attitudes and norms) cannot be used for targeting. However, our framework is flexible and can be applied in other settings where richer individual-level data is available. Third, compared to linear interaction models, causal forests are limited in terms of interpretability. To address this, we apply a comprehensive analytical framework to capture heterogeneity in the treatment effect estimates of the causal forest and identify key drivers of heterogeneity in the effectiveness of behavioral climate interventions.

Our work demonstrates how machine learning can uncover for whom and under what conditions climate change interventions are effective. By identifying individual-level differences in response, our framework contributes to theory building in climate communication and supports the development of personalized targeting strategies. Our findings challenge the traditional one-size-fits-all approach as commonly used in practice and instead show that tailoring interventions to individuals is crucial for effectively promoting belief in climate change. Building on these insights, researchers, policy-makers, and practitioners can design more effective, personalized communication strategies that better engage a broad range of audiences, including marginalized and other underrepresented groups. Importantly, our findings reveal that different types of interventions resonate depending on factors such as social norms and regional context. This suggests that communication strategies must be adapted to local conditions—especially in the Global South and in countries facing high climate risk—rather than relying on approaches developed primarily for the Global North.

4 Methods

Our framework for understanding the heterogeneous effectiveness of behavioral climate interventions across sociodemographic, attitudinal, and normative factors has three key steps. *First*, we analyzed the variation in effectiveness across behavioral interventions using data from a large-scale randomized controlled trial (RCT 1), where we applied the causal forest as a state-of-the-art machine learning method to estimate HTEs. This shows that the ATE often obscures meaningful individual-level differences. *Second*, we identified key drivers of this heterogeneity by examining how the differential effectiveness of interventions varies across covariates. Here, we start with sociodemographic characteristics (from RCT 1 due to the larger power), and further extend the analysis to include attitudinal and normative factors (from RCT 2 due to the more comprehensive set of covariates). *Third*, we tested the practical relevance of our findings. For this, we developed a personalized targeting strategy for climate-related communication that we evaluated using an online advertising platform. We then compared a personalized strategy (where we match individuals to interventions based on their estimated HTEs) against a non-personalized approach.

4.1 Data

RCT 1. We build on data from the International Climate Psychology Collaboration (ICPC), a large-scale RCT conducted across 63 countries with $N = 59,440$ participants [18, 20]. All 11 interventions used in RCT 1 were developed by behavioral scientists and pre-screened for cross-cultural feasibility and relevance to climate attitudes (see [18] for more details). After giving informed consent, participants were randomly assigned to one of 11 expert-crowdsourced behavioral interventions designed to improve climate change beliefs or a control condition involving a neutral task, i.e., reading a short excerpt from a fictional novel unrelated to climate change. Each intervention was purposefully designed to target a distinct psychological mechanism. For example, PSYCHDISTANCE frames climate change as a proximal risk; FUTURESELFCONT emphasizes the

future self-continuity regarding climate change; COLLECTACTION highlights the unique potentials of human collaboration to tackle climate change (see Supplementary Table S1 for a full list of the interventions). Participants then rated their stance towards climate change, which served as the outcome variable. Lastly, all participants provided a set of sociodemographic variables.

To measure climate change belief, participants were asked to rate four statements on a scale from 0 to 100: (1) “Taking action to fight climate change is necessary to avoid a global catastrophe”; (2) “Human activities are causing climate change”; (3) “Climate change poses a serious threat to humanity”; (4) “Climate change is a global emergency”. The final measure for the outcome climate change belief was computed as the average of those four items (Cronbach’s $\alpha = 0.934$; see [18]).

RCT 2. To enrich the set of individual-level covariates, we conducted another, preregistered online experiment (*RCT 2*: $N = 1,610$) on *Prolific*. Preregistration is available via <https://osf.io/h7mnc/>. Ethics approval was granted by the Ethics Committee at the Faculty of Management, Economics and Social Sciences at the University of Cologne (Reference: 240090DS). The experimental design of *RCT 2* was similar to that of *RCT 1*, but we extended it by asking participants an additional set of survey items regarding their attitudes and norms. Participants were randomly assigned to one of three interventions or a control condition. The interventions used in *RCT 2* were the three best-performing interventions from *RCT 1* based on the estimated ATEs, namely, PSYCHDISTANCE, COLLECTACTION, and FUTURESELFCONT. All participants gave informed consent at the beginning of the study and had to meet the following criteria: be at least 18 years old, have an approval rating of at least 95%, reside in the U.S., and have English as their first language. Participants were compensated at an hourly rate of \$8.47.

4.2 HTE estimation using the causal forest

To estimate HTEs, we applied the causal forest [38, 39], a nonparametric causal machine learning method specifically designed to capture heterogeneity in treatment effects [64] (see details in

Supplement D). The causal forest extends the traditional random forest [65] to enable valid causal inference. For this, the causal forest constructs trees that partition the data to maximize treatment effect heterogeneity. Hence, different from the traditional random forest, the outputs of the causal forest are the estimated HTEs conditional on some covariates (e.g., sociodemographic variables). Further, the causal forest uses an ensemble of many individual causal trees and then averages the estimates over the ensemble, which helps to reduce variance and improve generalization. Further methodological details are provided in Supplement D.

Formally, our estimand is the HTE, defined as $\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$, where Y represents climate change belief under treatment (i.e., $Y(1)$) versus control (i.e., $Y(0)$), and X denotes a set of observed variables, such as sociodemographic characteristics, attitudes, and norms. The HTE is often referred to as the conditional average treatment effect (CATE), because it captures individual-level variation in the response to an intervention *conditional* on the covariates. This distinguishes the HTE from the ATE, $\mathbb{E}[Y(1) - Y(0)]$, because the ATE provides only a single summary estimate of the overall impact of a treatment across the entire population. In contrast, the HTE allows us to analyze how intervention effects differ across subpopulations. For example, we can estimate $\tau(x)$ for individuals in countries with lower education, larger climate risk, and lower CO₂ per capita (such as encountered in the Global South) to understand how individuals from different regional and cultural contexts respond to climate interventions.

We chose the causal forest approach due to several key advantages. First, it does not rely on restrictive parametric assumptions and thus allows it to flexibly model complex, nonlinear interactions between variables and HTEs [38, 39]. Second, the causal forest can incorporate a high-dimensional set of variables to estimate HTEs, making it particularly well-suited to account for various effect modifiers that moderate the treatment effect heterogeneity. Third, the causal forest yields estimates that generalize well to out-of-sample data. This is achieved through (i) the ensemble structure, which stabilizes estimations by averaging across many randomized causal trees, and (ii) the use of “honesty” [38], a sample-splitting strategy that reduces bias in the estimation of

treatment effects.

4.3 Step 1: Heterogeneity in the effectiveness of behavioral climate interventions

4.3.1 Materials

We used RCT 1 to analyze the heterogeneity in the effectiveness of behavioral interventions on climate change belief. We intentionally focus on RCT 1 for this purpose due to the large sample size and thus statistical power. Climate change belief was chosen as the main outcome because it is likely to be a key predictor of downstream behaviors, such as support for climate action [40–43].

We used a comprehensive set of individual-level and country-level sociodemographic variables as predictors of heterogeneity (which were all collected in RCT 1). The variables were chosen to capture heterogeneity in how individuals respond to the behavioral climate interventions and, informed by earlier work [30], already played a crucial role in predicting baseline climate change belief. The individual-level sociodemographic variables include: (i) gender, (ii) income, (iii) education, (iv) age, (v) MacArthur scale of subjective social status [66], and (vi) political ideology. Country-level sociodemographics were computed based on the country the individuals live in and include: (i) the climate change risk indicator from the World Bank, (ii) the Gini index, (iii) human development index, and (iv) CO₂ emissions per capita. Details on the sociodemographic variables are provided in Table S2.

Data from four countries were excluded from the analysis due to missing values on country-level sociodemographic variables. Specifically, Taiwan was excluded due to missing data on the climate risk index, HDI, and Gini index. Further, Singapore, Saudi Arabia, and New Zealand were excluded due to missing data on the Gini index. In total, 2,200 individuals from these countries were excluded (reducing the sample size from $N = 59,508$ to $N = 57,308$).

4.3.2 Analysis

For each intervention, we used a separate causal forest to estimate HTEs by comparing individuals assigned to the intervention with those in the control group. The data were randomly split into a 70% training set and a 30% test set. Each causal forest was trained on the same training set and evaluated on the held-out test set. We applied best practices in machine learning and reported our results using only the test set. Hyperparameter tuning was performed using cross-validated out-of-bag error. All models were implemented using the `grf` package in R [67]. For more implementation details on model training and tuning, see Supplement D.

We performed model calibration of the HTE estimates of the causal forest following best practices [48]. Specifically, we performed Platt scaling [68] and centering of the mean estimation of HTEs in the training data to improve model calibration, with these same scaling values applied to the estimations for the test set. Model calibration refers to aligning estimations with true values, i.e., recalibrating estimations. However, with HTEs, the true difference in outcome between the treated and untreated states cannot be observed. Therefore, we used the observed ATE across tiles of estimated HTEs on the training data as our response. We then fitted a logistic regression with a quasi-binomial link, estimating the ATEs from the HTEs. The same logistic regression model was then used to estimate the calibrated HTEs.

To compare the HTEs against the ATE, we estimated the ATE and corresponding confidence intervals using the augmented inverse probability weighting (AIPW) estimator [45,46]. The AIPW estimator is a doubly robust, nonparametric method that is known to be both flexible in capturing nonlinear relationships and interactions and robust to skewed, non-normal, and bounded outcome distributions (which are both characteristics of our data). For this, the AIPW estimator combines propensity score weighting with outcome regression models to produce statistically efficient and unbiased estimates of the ATE [45,46]. Further methodological details are provided in Supplement D. Given the experimental design, the propensity score was known and fixed at 0.5, ensuring

the consistency of the estimator even in cases of potential misspecification of the outcome models. By incorporating variables into the outcome regression, the AIPW estimator also achieves reduced variance as compared to a simple difference-in-means approach. To avoid overfitting and ensure unbiased estimation, we applied cross-fitting, that is, the training data were split into two folds, with the outcome models trained on one subset and ATEs estimated on the other. We implemented the AIPW estimator using the `grf` package (version 2.3.2) [67] in R (version 4.2.1).

4.3.3 Robustness checks

We conducted several robustness checks of the estimated HTEs from our causal forest model. We first applied the effect scores method [47–49], following best practices in causal machine learning. For each intervention, the trained causal forest was applied to the unseen test set, after which individuals were sorted into quartiles based on their estimated HTEs. Within each quartile, we estimated the ATE using AIPW [45, 46] and then visually inspected whether the ordering of the quartile ATEs aligned with the ordering of the estimated HTEs. As a second robustness check, we compared the HTE estimates from our causal forest model to those from a T-learner [50], which estimates separate outcome models for the treatment and control groups and computes HTEs as their difference. For each intervention, we computed Pearson’s correlation coefficient between the HTE estimates from the causal forest and the T-learner, which showed that the results are consistent across different estimation strategies.

4.4 Step 2: Drivers of heterogeneity

4.4.1 Materials

Here, we aimed to identify the drivers of heterogeneity in the effectiveness of behavioral climate interventions. For this, we proceeded in two steps: (1) To understand heterogeneity across sociodemographics, we analyzed data from RCT 1. This allowed us to identify the underlying sociodemographic drivers of heterogeneity, namely the individual-level and country-level sociodemographics

described above. (2) To understand heterogeneity across attitudinal and normative drivers, we performed RCT 2 to obtain additional data and thus identify attitudinal and normative drivers of heterogeneity. The additionally collected variables were the following: (i) perceived scientific consensus, (ii) trust in climate science (iii) trust in government, (iv) identity with human welfare, (v) identity with global citizens, (vi) environmental identity, (vii) external and, (viii) internal environmental motivation, (ix) pluralistic ignorance, (x) and New Ecological Paradigm (NEP) survey. Details about the variables are provided in Table S3.

4.4.2 Analysis

We analyzed the drivers of heterogeneity by examining how the effectiveness of behavioral climate interventions varied across different variables. The main purpose was to identify which variables systematically moderated the estimated HTEs, thereby serving as heterogeneity drivers. We structured the analysis based on two sets of variables: (1) sociodemographic variables (where we reanalyzed data from RCT 1) and (2) attitudinal and normative variables (using the data from the newly collected RCT 2).

To assess whether the collected variables systematically explain heterogeneity in treatment effects, we analyzed the distribution of each variable across tertiles of individuals sorted by their estimated HTEs. For each intervention, individuals were grouped into lower, middle, and upper tertiles based on their HTEs estimated via causal forests. We then compared the distributions of each variable across these HTE tertiles. To formally test for differences between tertiles, we applied the non-parametric Kruskal-Wallis test [53], which is well-suited for detecting differences in distributions without the normality assumption. All variables were standardized to enable comparability across measurement scales, and we reported standardized means and standard errors across tertiles. This procedure allowed us to identify which variables drive heterogeneity in intervention effectiveness.

We further examined which interventions are most effective for which subgroups. Specifically,

we analyzed the proportions of individuals in each subgroup for whom a given intervention was estimated to be optimal, where subgroups were defined by the sociodemographics, attitudes, and norms. For this, we first determined which of the interventions yields the highest estimated HTE for every individual. We then split the different variables into subgroups, either based on the pre-defined categories from [18] or by splitting continuous variables into three equally sized subgroups (e.g., low, middle, and high income). For age, we opted for subgroups that are consistent with the categorization on *Google Ads* (i.e., 18–24, 25–44, 45–64, and 65+). Afterward, we calculated the proportion of individuals within each subgroup for whom a given intervention is optimal. Here, we again estimated HTEs on a held-out test set. By comparing the distribution of optimal interventions across subgroups, we can assess, on the one hand, whether a variable moderates effectiveness, and on the other hand, which intervention is best suited for which population subgroup.

4.4.3 Robustness checks

We conducted an additional analysis to assess the robustness of our findings. Specifically, we computed SHAP values [55] to explain the contributions of each variable to the HTE estimates produced by our causal forest (see Supplementary Figure S10 and S11).

4.5 Step 3: Online platform study to show ecological validity

4.5.1 Materials

We tested the ecological validity of our results using a preregistered online platform study on *Google Ads*. The preregistration is available via <https://osf.io/h7mnc/>. Ethics approval was granted by the Ethics Committee at the Faculty of Management, Economics and Social Sciences, University of Cologne (Reference: 240090DS). We then tested whether our previous analyses could inform a personalized targeting strategy that performs better in engagement with ads (measured through clicks) in comparison to a non-personalized approach. The design of our online platform study followed previous work that used *Google Ads* in other contexts [69–71]. We conducted our

online platform study using a U.S. sample of English-speaking individuals and focused on demographic variables that were available for targeting within the *Google Ads* platform, namely gender and age. Note that other variables are not available for targeting on *Google Ads* due to privacy reasons. We designed the ads similarly to how the original interventions were designed in RCT 1 [18] in terms of textual and visual cues (see Supplementary Figures S13 to S15). We conducted pretests to ensure that the ads target the psychological mechanisms of the original interventions. We further re-estimated the causal forest using only the U.S. subsample from RCT 1 for better comparability and identified the best-performing interventions per subgroup.

Following the predefined categories on *Google Ads* (two for gender and four for age), we specified eight subgroups. Within each subgroup, we aggregated the HTEs of all individuals to subgroup ATEs using the doubly robust AIPW estimator [45, 46]. Finally, for each subgroup, the intervention with the highest subgroup ATE informed our personalized approach (see Supplementary Figures S12). Hence, a user was randomly assigned to one of the following two approaches: (1) the non-personalized approach, in which users were presented with one of the three top-performing interventions (based on the ATE), or (2) the personalized approach, in which users were presented with the best-performing intervention based on the highest estimated HTE for the corresponding subgroup. All behavioral interventions were operationalized as advertisements consisting of an image and a short description (see Supplementary Figures S13 to S15).

We specified the following settings within *Google Ads*, following best practice [71], to ensure comparability between the two conditions for a given sociodemographic group. First, we designed the ad campaigns such that they were available for two full consecutive days (from August 20 to August 21). Second, we specified an overall budget of €1,120 for both conditions, which was evenly split between campaigns to ensure full comparability on the campaign level. Once the budget for a campaign was spent for the day, the campaign continued to be available the next day. Third, we leveraged the internal A/B testing environment from *Google Ads* to compare campaigns fairly and prevent cross-condition spillover, which ensured that each user could only see one con-

dition. Fourth, we disabled the internal optimization of *Google Ads*, which otherwise promotes the best-performing ads more heavily to maximize the CTR, so that no intervention gained an unintended advantage. Fifth, we turned off settings that could cause users to see the same ad more than once (i.e., frequency capping was configured to limit each ad to a single impression per user). Lastly, we excluded mobile devices from our campaigns due to high variability in user behavior (e.g., to avoid unintended clicks on small touch devices).

4.5.2 Analysis

To evaluate the effectiveness of our personalized approach compared to a non-personalized approach, we employed logistic regression with click behavior as the binary outcome variable. The model specification compared click rates between users who received their estimated, best-performing intervention (personalized approach) versus those who received a randomly assigned intervention (non-personalized approach). The logistic regression equation is given by

$$\text{logit}(P(\text{Click}_i = 1)) = \beta_0 + \beta_1 \cdot \text{Personalized}_i, \quad (1)$$

where Click_i represents the binary click outcome for individual i , Personalized_i is an indicator variable for receiving the personalized approach based on demographic subgroup assignment, and β_1 quantifies the log-odds increase associated with our personalized approach. The coefficient β_1 thus captures the effectiveness of personalized targeting vs. non-personalized targeting. To estimate the logistic regression, we used the function `glm` from R (version 4.2.1).

4.6 Extension to climate policy support

We repeated our entire analysis using a different outcome, namely, climate policy support (see Supplementary Figures S16 to S20). This outcome consisted of the level of agreement of participants with the following nine statements: (1) “I support raising carbon taxes on gas/fossil fuels/coal,”

(2) “I support significantly expanding infrastructure for public transportation,” (3) “I support increasing the number of charging stations for electric vehicles,” (4) “I support increasing the use of sustainable energy such as wind and solar energy,” (5) “I support increasing taxes on airline companies to offset carbon emissions,” (6) “I support protecting forested and land areas,” (7) “I support investing more in green jobs and businesses,” (8) “I support introducing laws to keep waterways and oceans clean,” and (9) “I support increasing taxes on carbon intense foods (for example meat and dairy).” The agreement was measured on a Likert scale from 0 (= not at all) to 100 (= very much so) and subsequently averaged across the different dimensions to obtain an overall score.

Similar to our main analysis, we first plotted the distribution of the HTE estimates across individuals and interventions (see Supplementary Figure S16). Next, we aimed to identify sociodemographic drivers (see Supplementary Figure S17) and attitudinal and normative drivers (see Supplementary Figure S18) of heterogeneity through a stratified analysis of the variables across HTE tertiles. Finally, we estimated the optimal interventions by sociodemographic subgroups (see Supplementary Figure S19) and attitudinal and normative subgroups (see Supplementary Figure S20). Overall, we again find substantial heterogeneity in the effectiveness of the different behavioral climate interventions on the outcome of climate policy support. Again, the best intervention (now COLLECTACTION) by ATE is effective for only 78%, but backfires for 22% of the individuals. Similarly, the PSYCHDISTANCE intervention is effective for 66% of the individuals but harmful for 34%. As before, PSYCHDISTANCE is more effective for females, as well as individuals from countries with high CO₂ emissions and high HDIs.

Data availability

The complete dataset for RCT 1 is available at <https://zenodo.org/records/10345806>. Our dataset for RCT 2 and the online platform study are available at <https://osf.io/h7mnc/>.

Data for the country-level sociodemographics are available as follows: the climate change

730 risk indicator was sourced from <https://prosperitydata360.worldbank.org/en/dataset/IMF+RDD>,
731 the Gini index from <https://data.worldbank.org/indicator/SI.POV.GINI>, human development in-
732 dex (HDI) [66] from <https://ourworldindata.org/human-development-index>, and CO₂ emissions
733 per capita from <https://ourworldindata.org/co2-emissions#per-capita-co2-emissions>.

734 **Code availability**

735 All code to replicate our analyses is available at <https://osf.io/h7mnc/>.

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Author contributions

A.M. implemented the causal ML framework. A.M. and S.F. designed the causal ML framework. S.D. collected RCT 2 and performed the online platform study. A.M., S.D., M.W., and S.F. designed the online platform study. A.M. and S.D. performed the analysis of the online platform study. A.M. prepared the first draft of the manuscript. All authors contributed to conceptualization, manuscript writing, and approved the manuscript.

Competing interests

The authors declare no competing interests.

Supplements

Supplement A Supplementary figures

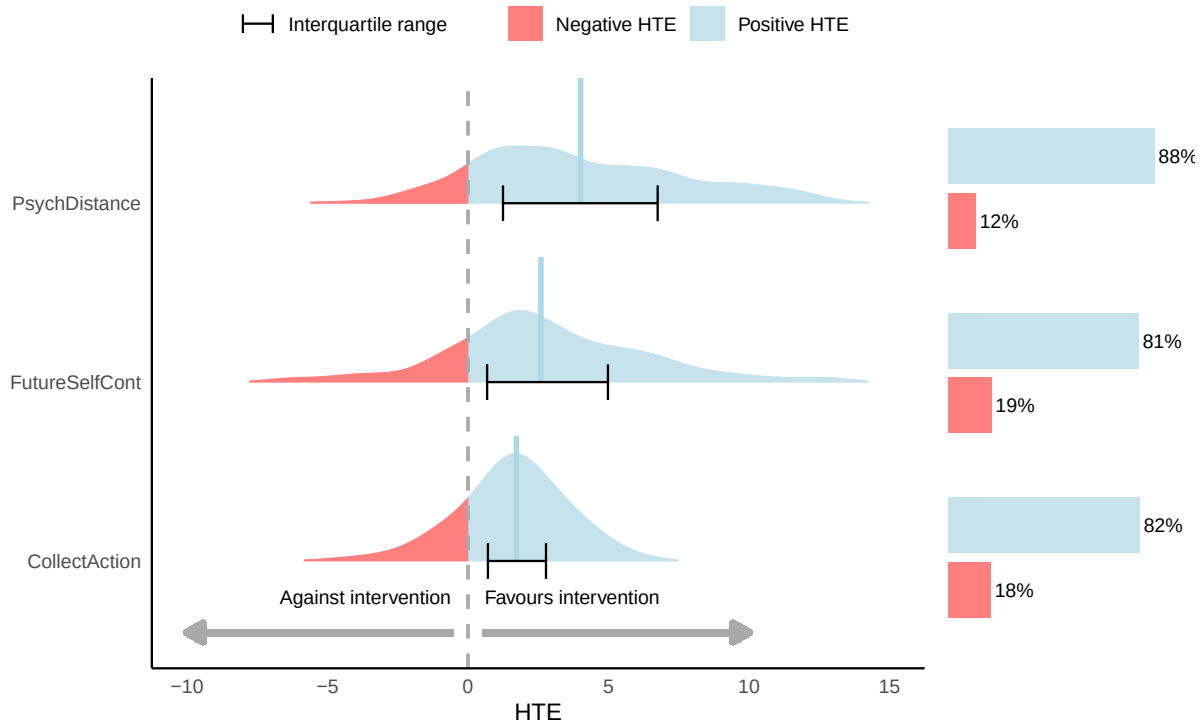
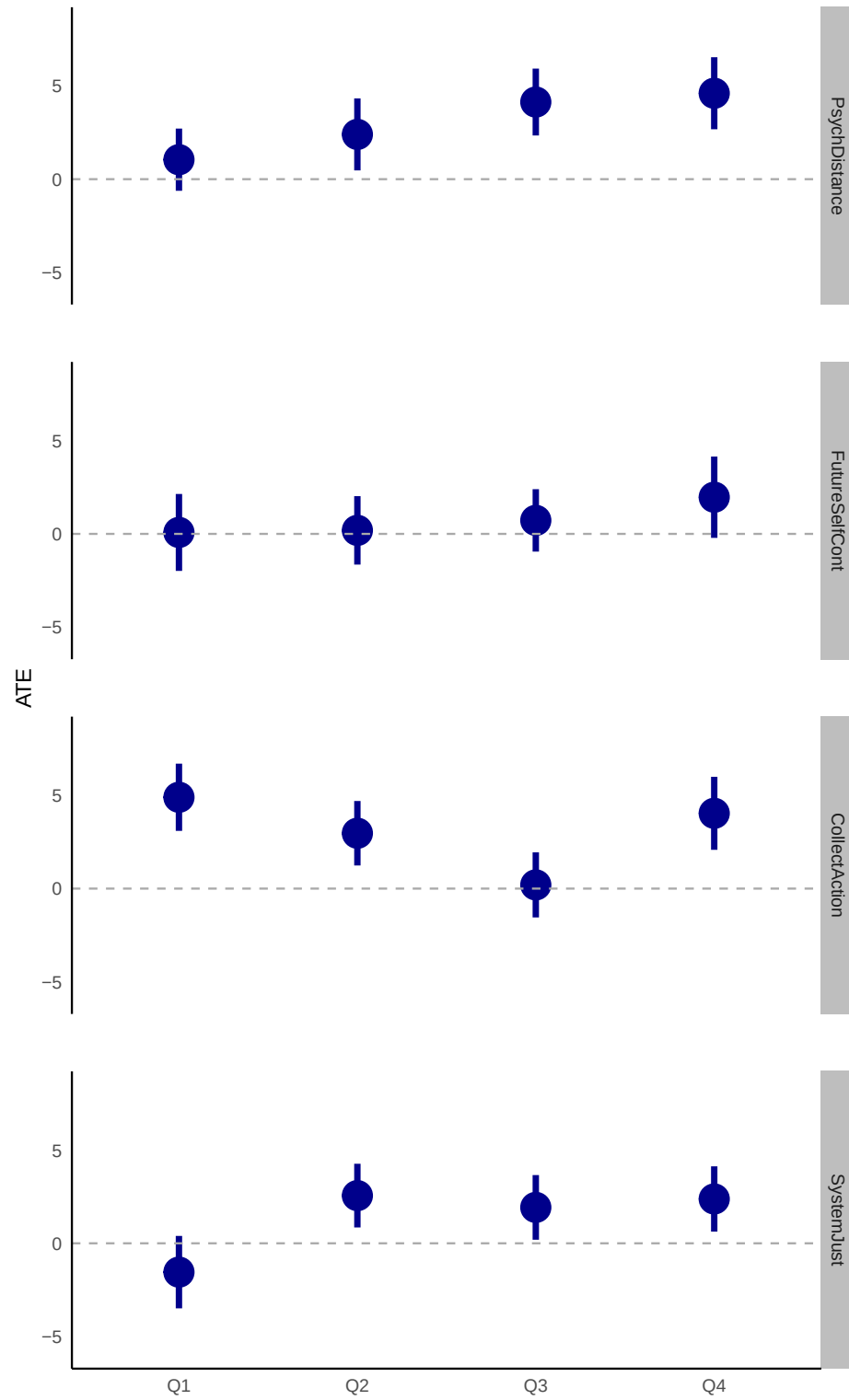
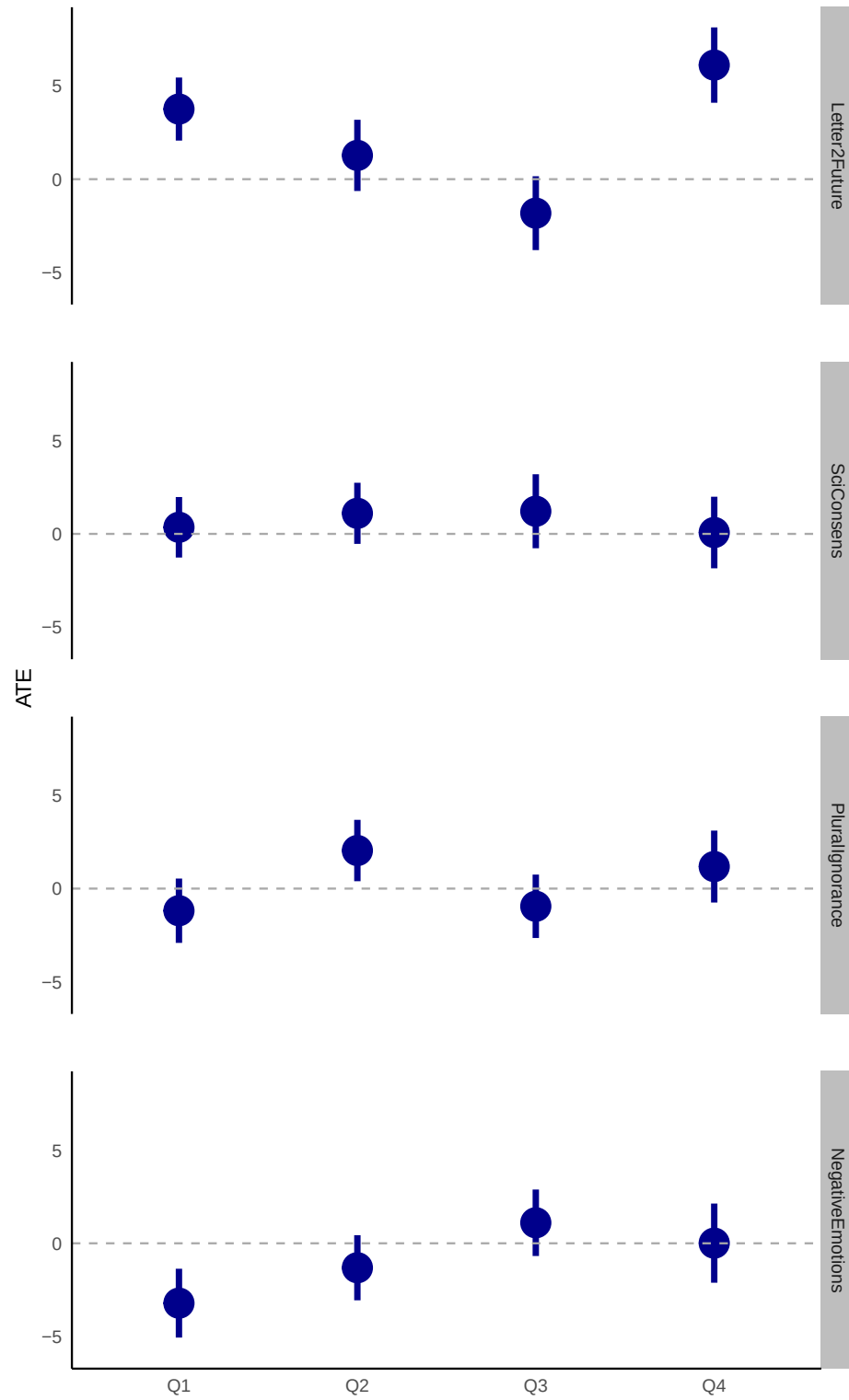


Figure S1: Estimated HTEs across interventions and individuals from RCT 2. The figure shows the distribution of estimated heterogeneous treatment effects (HTEs) for each of the three behavioral climate interventions, based on experimental data from RCT 2 ($N = 1,610$ U.S.-based participants). Each HTE represents the estimated difference in climate change belief compared to a control condition, conditional on individual variables. The bar charts indicate the proportion of individuals for whom an intervention is estimated to have a positive (blue) or negative (red) effect. For ease of interpretation, we indicate the interquartile range (horizontal whiskers) and the average treatment effect (ATE; vertical lines) estimated using a doubly robust augmented inverse propensity weighting (AIPW) method [45,46]. We note that the distributions of HTEs are expected to differ from those in Fig. 2 due to the usage of RCT 2, which included U.S.-based individuals and an extended set of variables (i.e., the attitudes and norms). For a full description of interventions, see Supplementary Table S1.





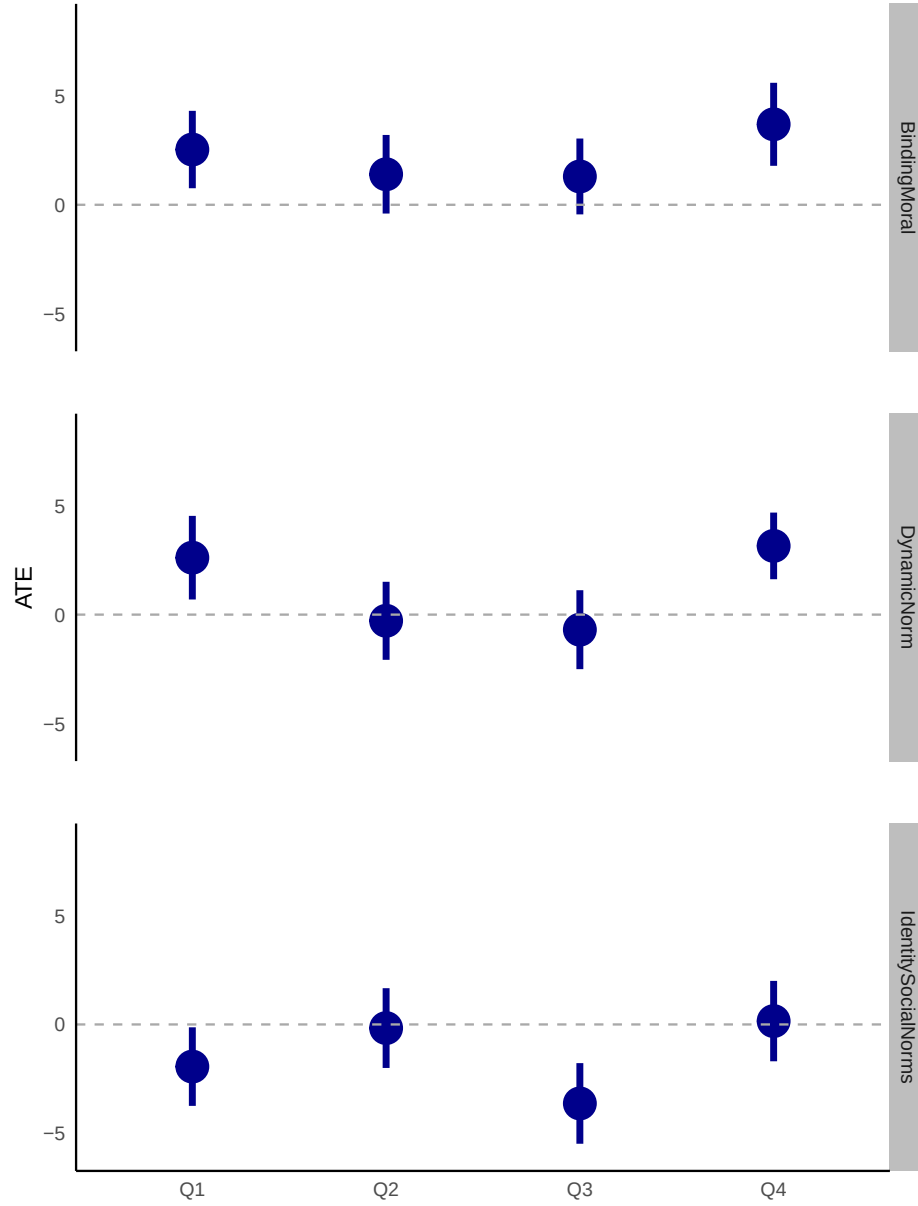


Figure S2: **External validation of the causal forest models.** The estimated HTEs for each intervention were estimated on an unseen test set, sorted in descending order, and split into quartiles. ATEs were estimated within each quartile using AIPW [45,46]. Overall, we observe an ascending ordering of the ATEs across quartiles for each intervention. This indicates that the causal forest models are robust. The whiskers represent the SEs.

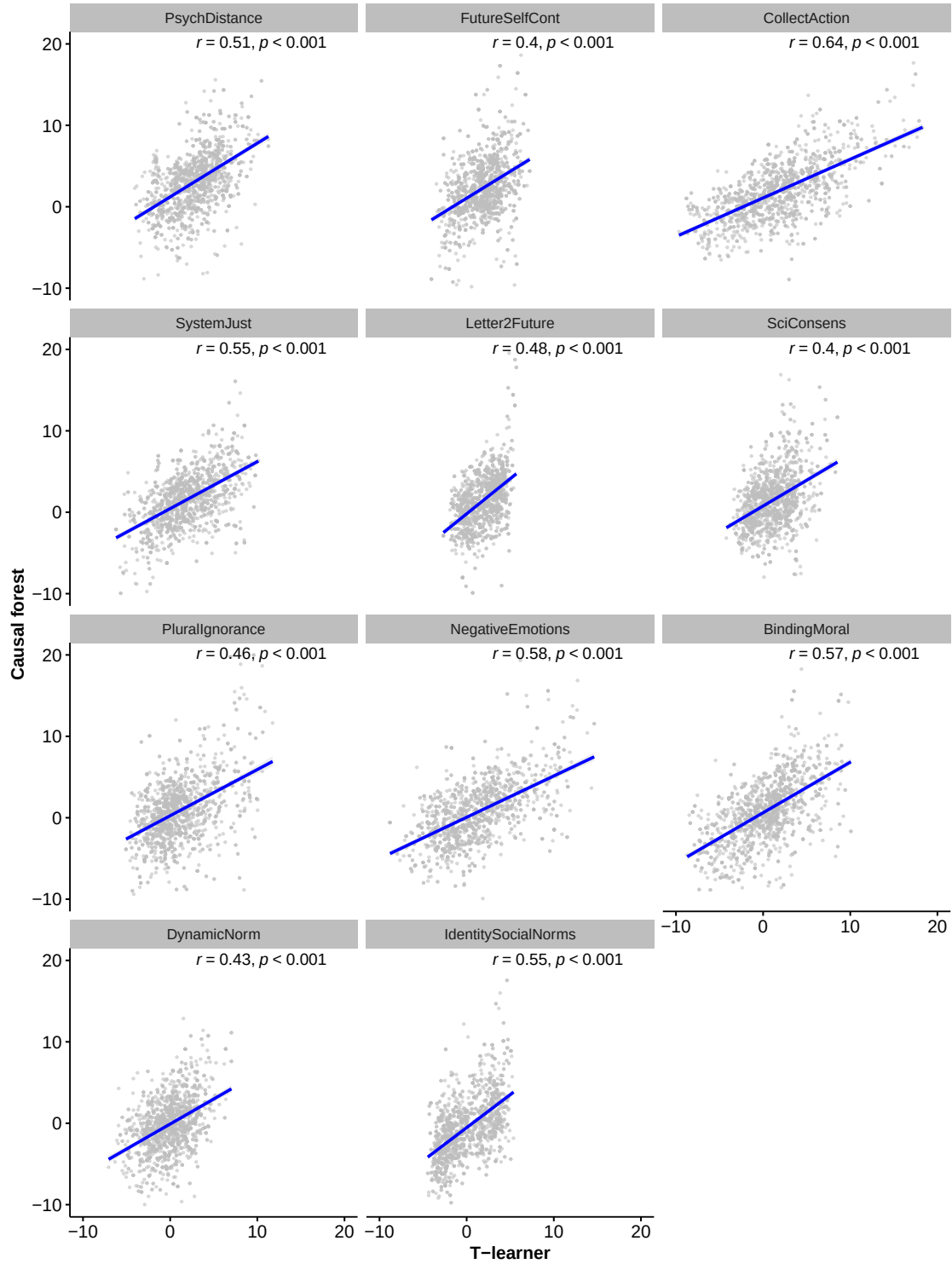


Figure S3: Correlation of the HTE estimates obtained from the causal forest model vs. the T-learner. As a robustness check, we estimated HTEs using the T-learner [50] (instantiated via a random forest) and compared them to the estimated HTEs from the causal forest model on the unseen test set for each intervention. We computed a linear regression line (blue) and Pearson's correlation coefficient. Overall, we found that the HTE estimates are strongly correlated (Pearson's correlation coefficient: $r > 0.4$, $p < 0.001$ for all interventions). This suggests that the HTE estimates are largely consistent across different estimation strategies.

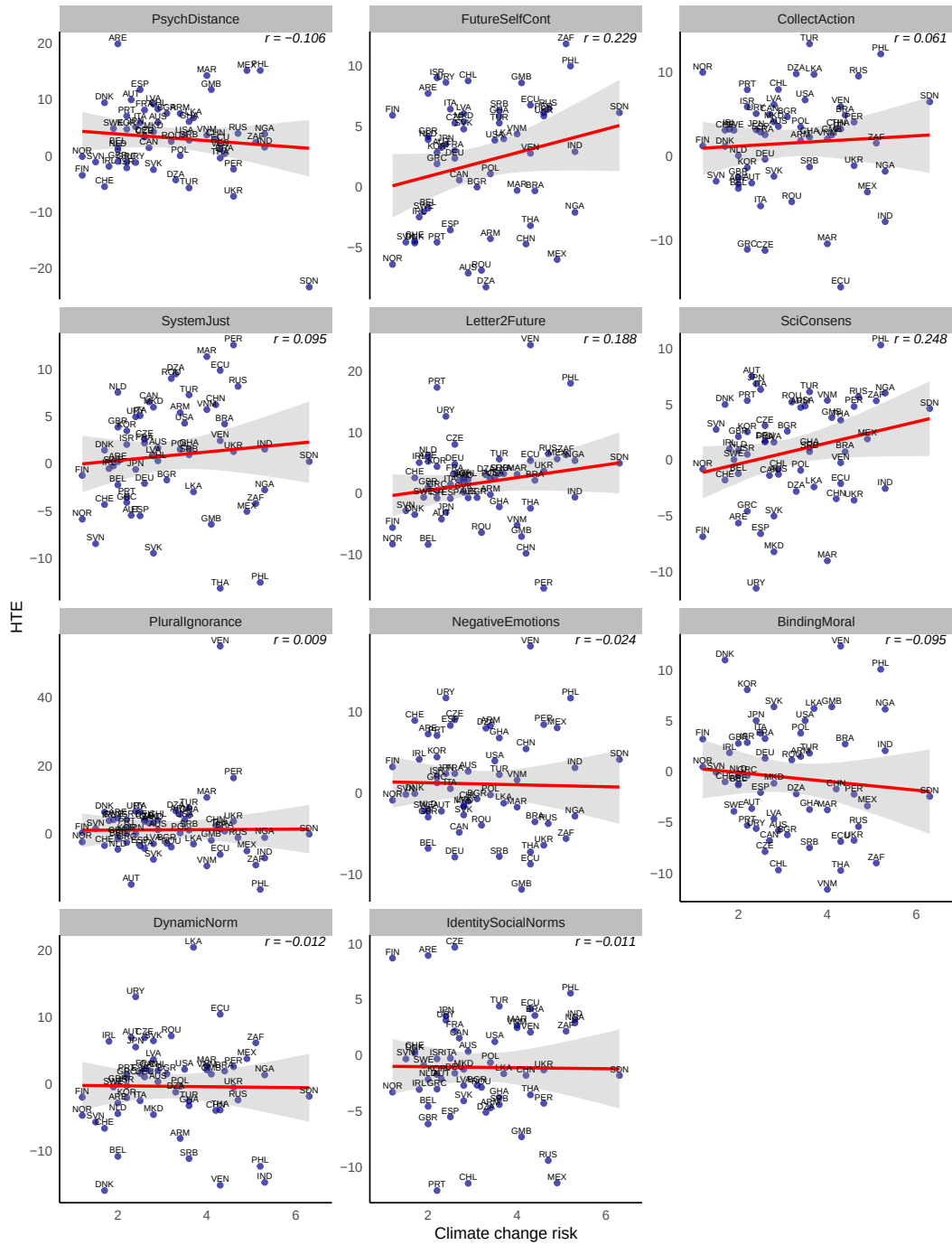


Figure S4: **Relationship of estimated HTEs and the corresponding climate change risk value for each country.** Each point represents the estimated HTE and the corresponding climate change risk value for a specific country (marked using ISO country codes). Subgroup HTEs were estimated using a doubly robust AIPW method [45,46]. We also compute linear regression lines (red), 95% confidence intervals, and Pearson's correlation coefficient.

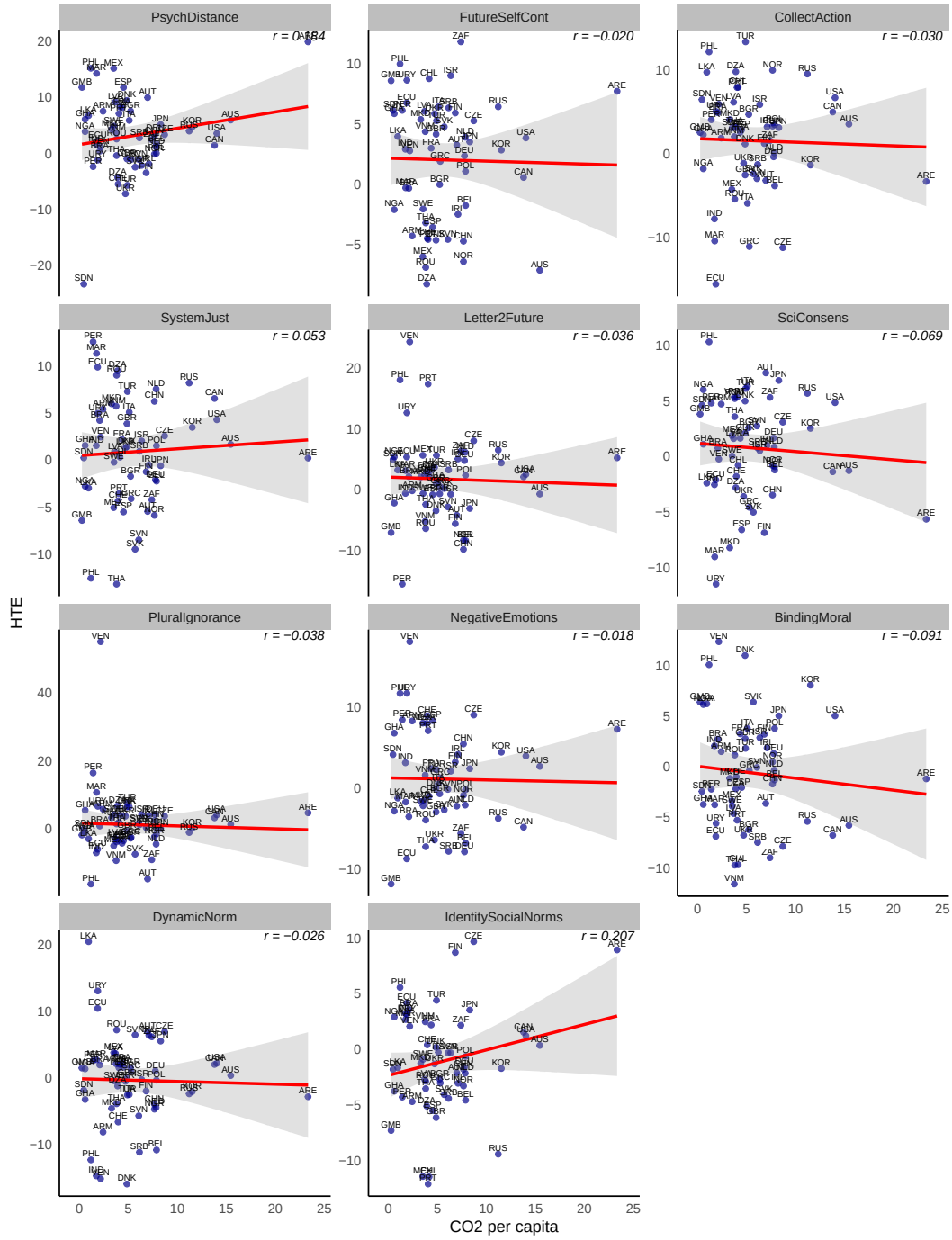


Figure S5: **Relationship of estimated HTEs and the corresponding CO₂ per capita for each country.** Each point represents the estimated HTE and the corresponding CO₂ per capita for a specific country (marked using ISO country codes). Subgroup HTEs were estimated using a doubly robust AIPW method [45,46]. We also compute linear regression lines (red), 95% confidence intervals, and Pearson's correlation coefficient.

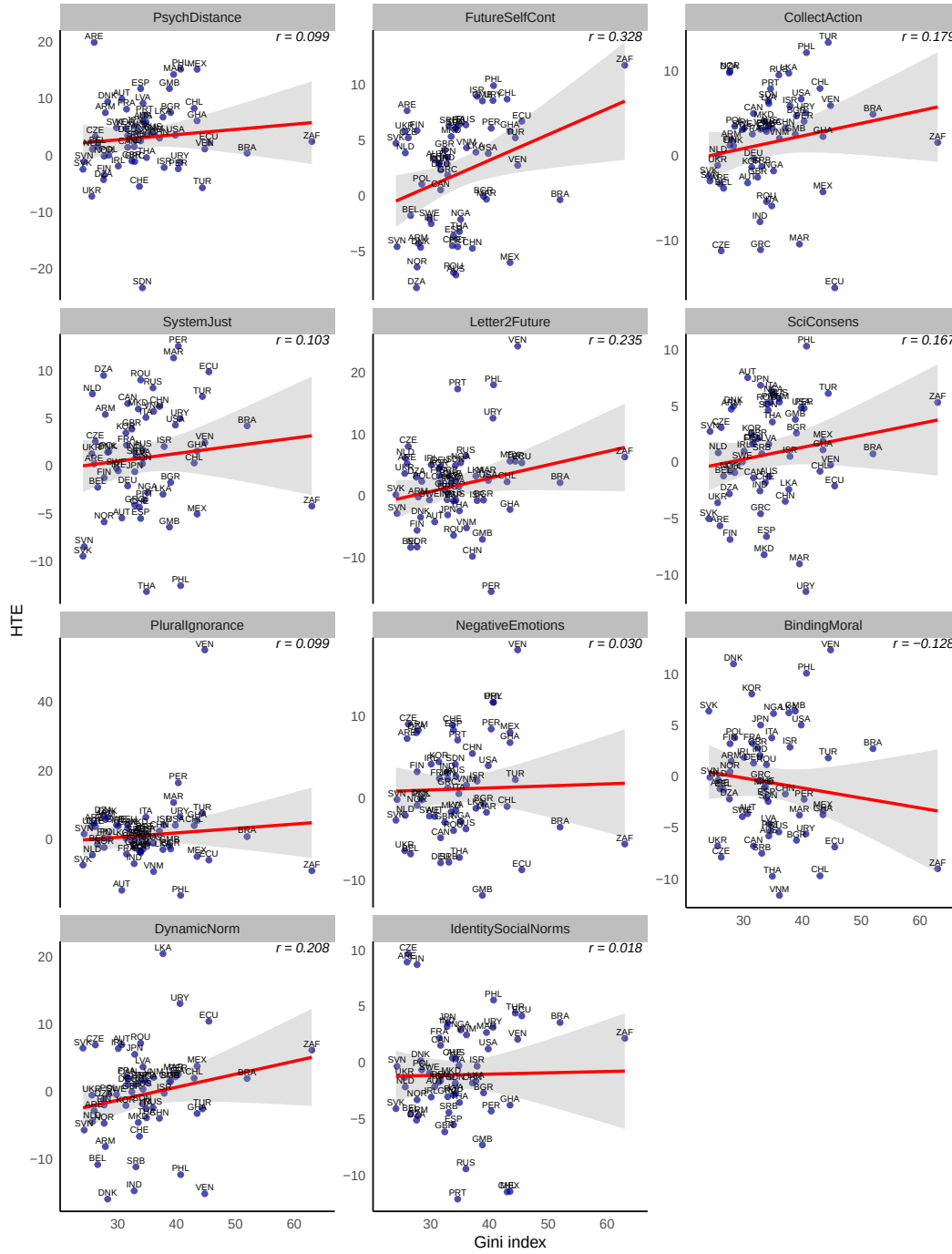


Figure S6: **Relationship of estimated HTEs and the corresponding Gini index for each country.** Each point represents the estimated HTE and the corresponding Gini index for a specific country (marked using ISO country codes). Soubgroup HTEs were estimated using a doubly robust AIPW method [45,46]. We also compute linear regression lines (red), 95% confidence intervals, and Pearson's correlation coefficient.

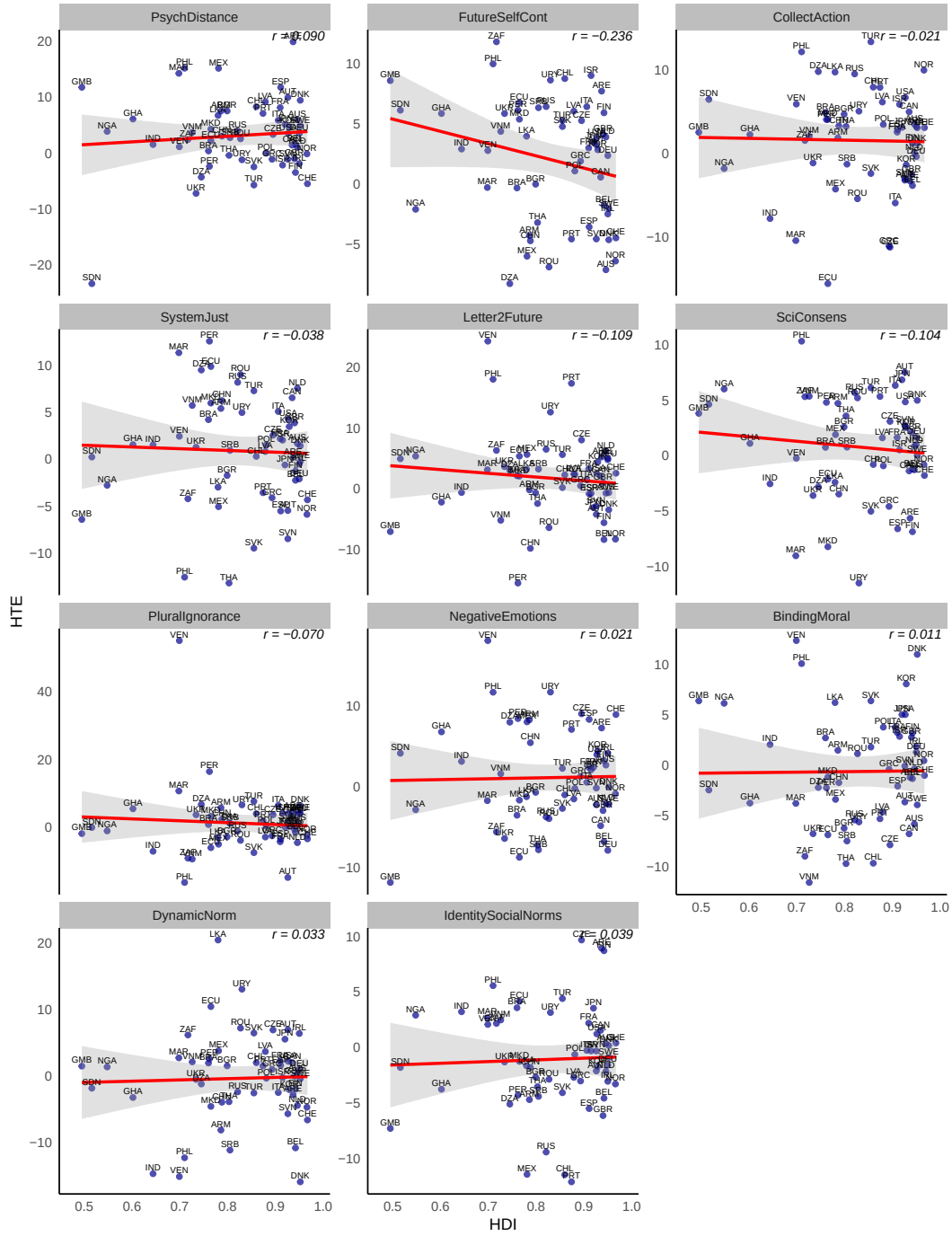


Figure S7: **Relationship of estimated HTEs and the corresponding HDI for each country.** Each point represents the estimated HTE and the corresponding HDI for a specific country (marked using ISO country codes). Subgroup HTEs were estimated using a doubly robust AIPW method [45,46]. We also compute linear regression lines (red), 95% confidence intervals, and Pearson's correlation coefficient.

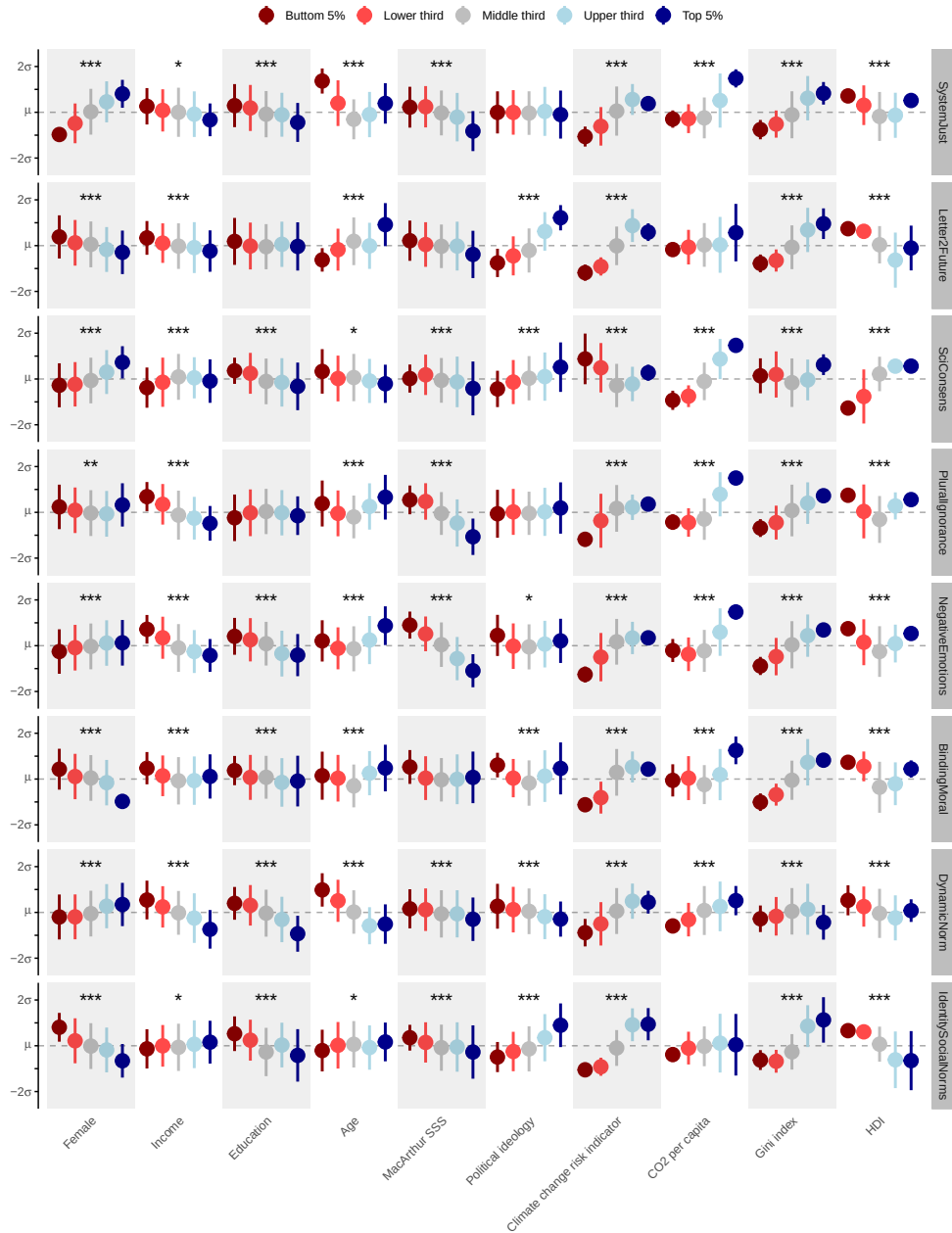


Figure S8: Sociodemographic drivers of heterogeneity in the effectiveness of the eight other behavioral climate interventions. Here, we assess whether there is a systematically higher intervention effectiveness for certain attitudes or norms. The plot shows how attitudes and norms, including variables referring to the environmental identity and the new ecological paradigm, are distributed across individuals. Here, we stratified the individuals into tertiles by the estimated HTEs for the eight interventions that were not covered in the main analysis. In addition to the tertiles, we also show the bottom and upper-5% tails of the estimated HTEs. Each point represents the standardized mean of a variable within a group, and the error bars show the standard error. Variables are standardized (mean = 0, SD = 1) to allow for comparability across scales. A point at $+1\sigma$ means that the average value of the variable in the specific tertile is one standard deviation higher than the overall population mean. Differences across tertiles are tested using the non-parametric Kruskal-Wallis test [53], with significance thresholds indicated as: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

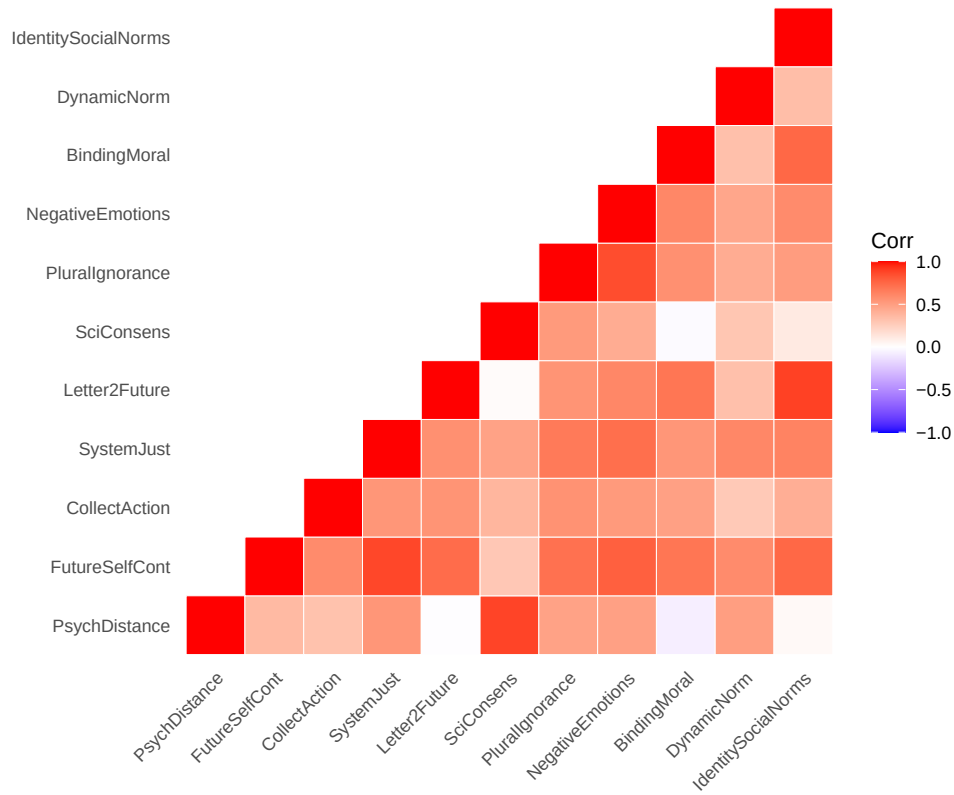


Figure S9: Correlations between the estimated HTEs of each intervention pair. The plot shows the correlations between the estimated HTEs of each intervention pair. Specifically, for each individual in the test set, we estimated the HTE of all eleven interventions. Then, we calculated Pearson’s correlation coefficient between the estimated HTEs of each intervention pair. Overall, the HTEs of most intervention pairs are positively correlated, and a few pairs are not or even negatively correlated (e.g., PSYCHDISTANCE and BINDINGMORAL). Hence, the effectiveness of some interventions appears to be shaped by similar variables, which suggests that the effectiveness may be due to similar psychological mechanisms, while other interventions show only small correlations, implying that different psychological mechanisms are responsible for the effectiveness.

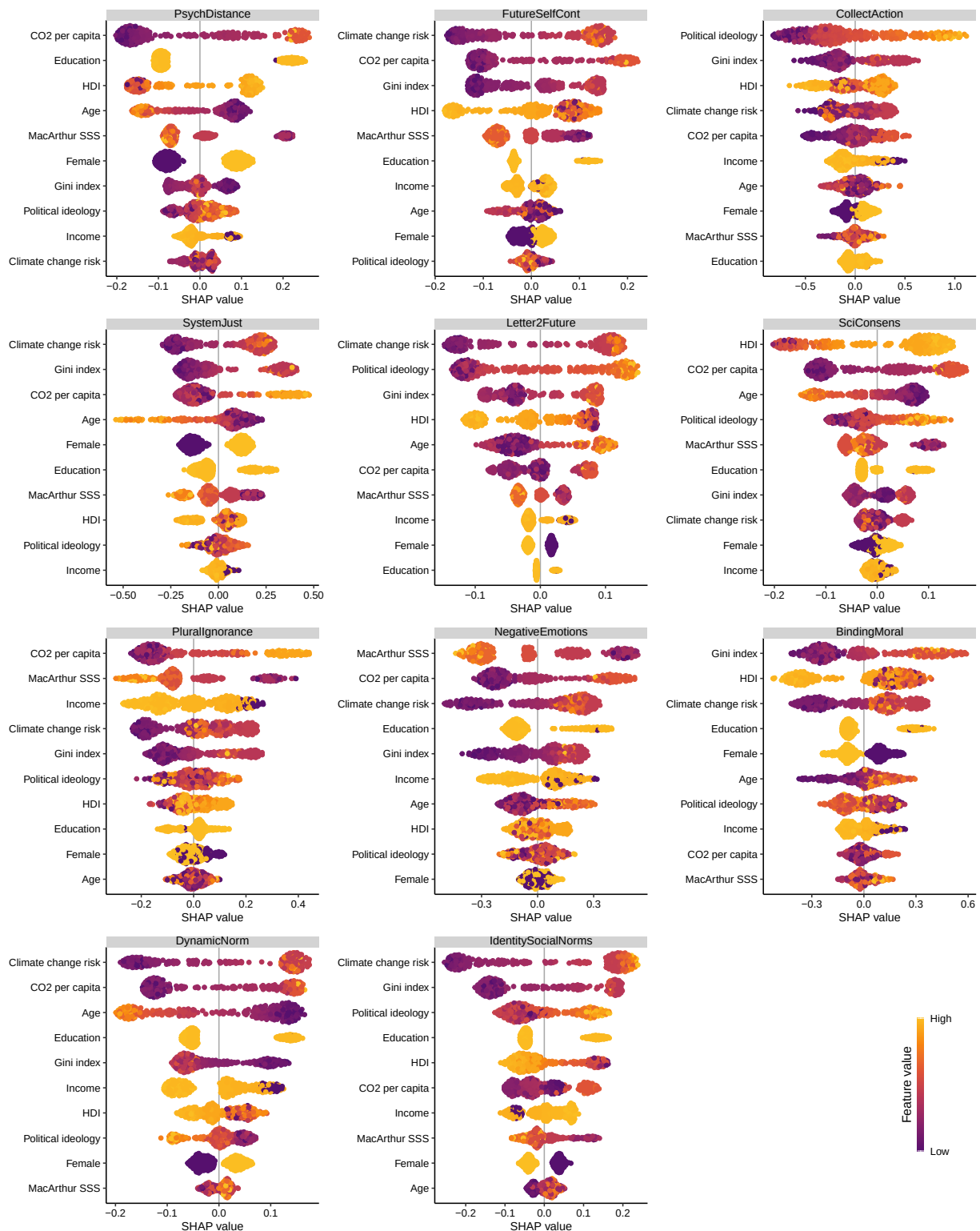


Figure S10: **SHAP value plots for all interventions in RCT 1.** Each dot represents a SHAP value [55] for a variable across different individuals, where the color indicates a high (yellow) to low (purple) SHAP value. A SHAP value of a variable measures its contribution to the estimation of an HTE relative to the average estimated HTE. Therefore, SHAP values explain drivers of heterogeneity that are associated with a larger (or smaller) HTE. Here, the variables are sorted based on their mean absolute SHAP values across the unseen test sets, so that more important determinants of HTEs are at the top.

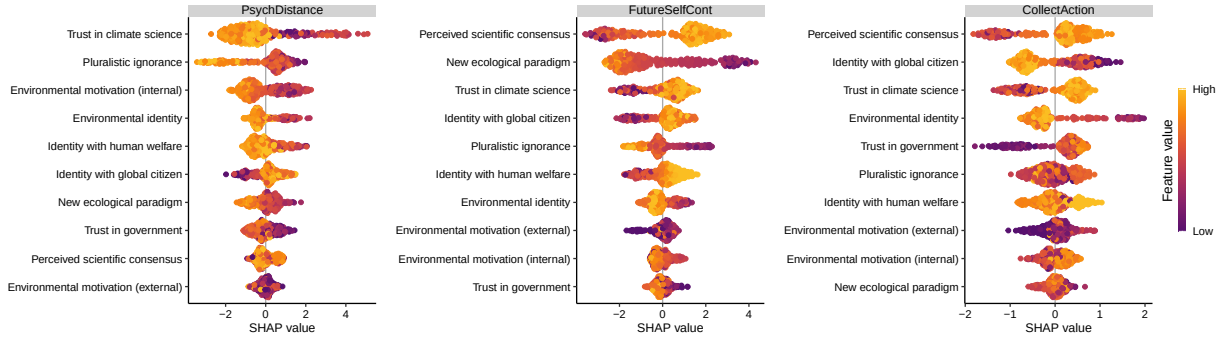


Figure S11: **SHAP value plots for all interventions in RCT 2.** Each dot represents a SHAP value [55] for a variable across different individuals, where the color indicates a high (yellow) to low (purple) SHAP value. A SHAP value of a variable measures its contribution to the estimation of an HTE relative to the average estimated HTE. Therefore, SHAP values explain drivers of heterogeneity that are associated with a larger (or smaller) HTE. Here, the variables are sorted based on their mean absolute SHAP values across the unseen test sets, so that more important determinants of HTEs are at the top.

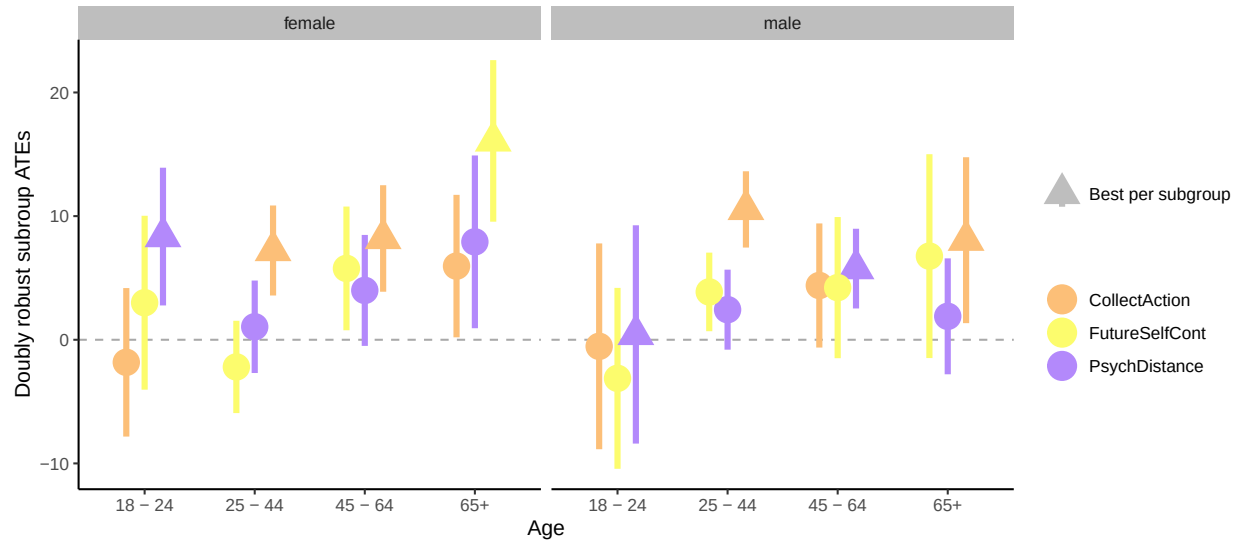


Figure S12: **ATEs for different subgroups of gender and age informing our personalized targeting approach.** Subgroup ATEs were estimated using a doubly robust AIPW method [45,46]. The best-performing intervention per subgroup is marked as a triangle and is assigned to that subgroup within our personalized targeting approach. Hence, for example, the analysis suggests that, for females aged 18-24, the PSYCHDISTANCE intervention may be most effective, so that we allocate the PSYCHDISTANCE for this subgroup in our personalized targeting strategy.



Figure S13: **COLLECTACTION** intervention as presented in our online platform study on *Google Ads*. The intervention was carefully constructed to resemble the same psychological mechanisms as presented in [18].



Figure S14: **FUTURESELFCONT** intervention as presented in our online platform study on *Google Ads*. The intervention was carefully constructed to resemble the same psychological mechanisms as presented in [18].



Figure S15: **PSYCHDISTANCE** intervention as presented in our online platform study on *Google Ads*. The intervention was carefully constructed to resemble the same psychological mechanisms as presented in [18].

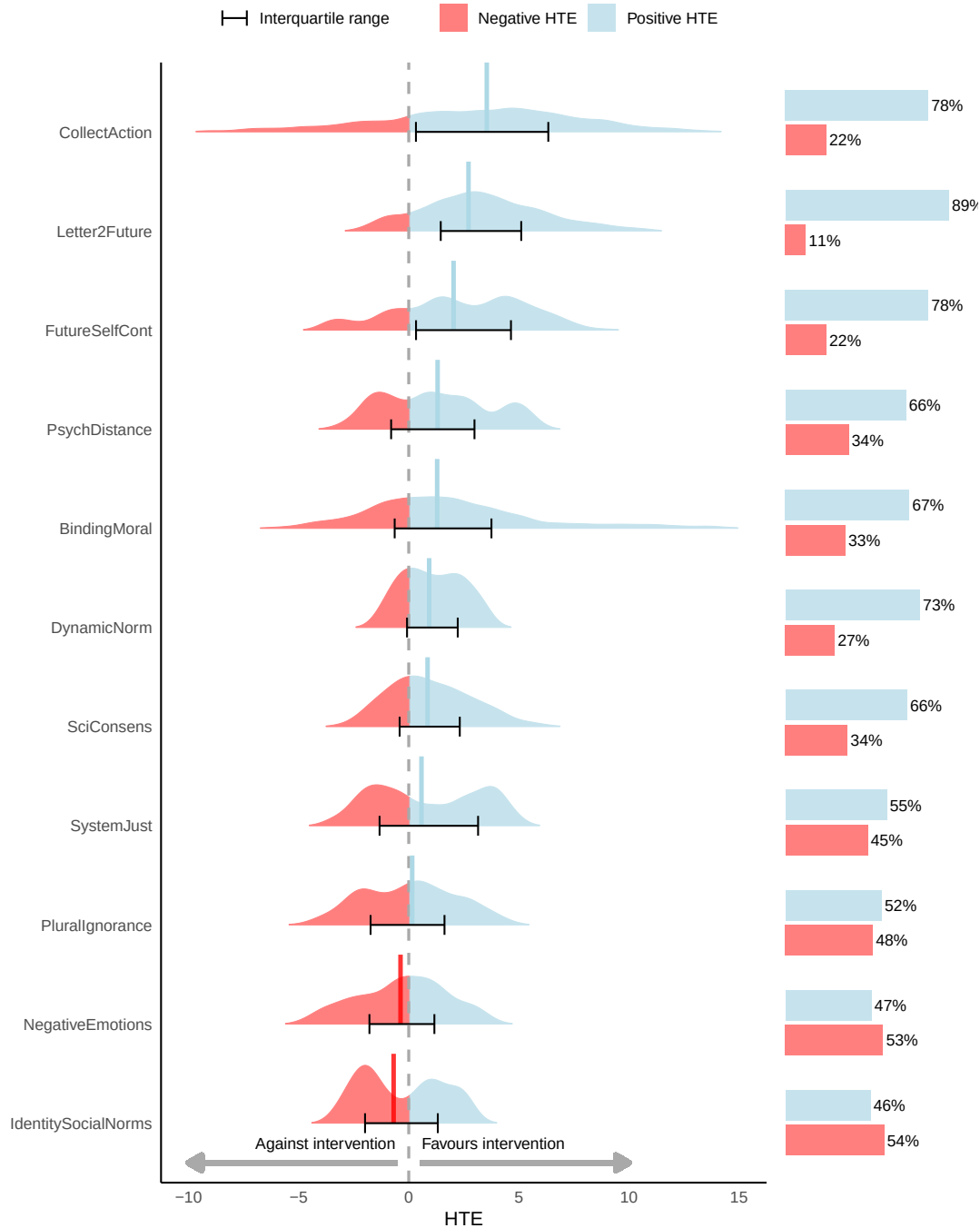


Figure S16: Estimated HTEs across interventions and individuals on the outcome climate policy support. The figure shows the distribution of estimated heterogeneous treatment effects (HTEs) on the outcome climate policy support for each of the 11 behavioral climate interventions, based on experimental data from RCT 1 ($N = 59,508$ participants across 63 countries). Each HTE represents the estimated difference in climate change belief compared to a control condition, conditional on individual variables. The bar charts indicate the proportion of individuals for whom an intervention is estimated to have a positive (blue) or negative (red) effect. For ease of interpretation, we indicate the interquartile range (horizontal lines) and the average treatment effect (ATE) estimated using a doubly robust augmented inverse propensity weighting (AIPW) method [45,46]. For a full description of interventions, see Supplementary Table S1.

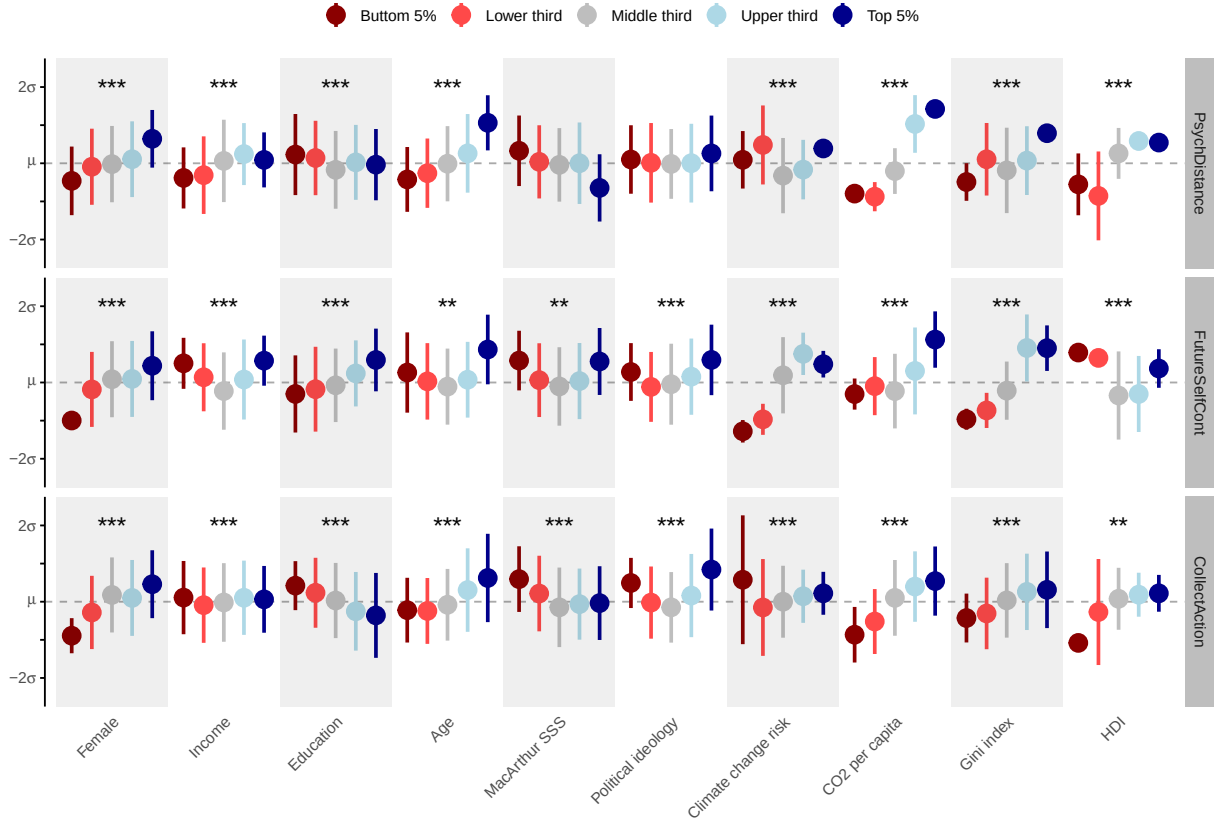


Figure S17: Sociodemographic drivers of heterogeneity in the effectiveness of behavioral climate interventions on the outcome climate policy support. Here, we assess whether certain characteristics are systematically associated with higher intervention effectiveness on climate policy support. The plot shows how the sociodemographic variables, including the MacArthur scale of subjective social status (MacArthur SSS) and the human development index (HDI), are distributed across individuals with different estimated effects of behavioral climate interventions. Here, we stratified the individuals into tertiles by the estimated HTEs for the three selected interventions: PSYCHDISTANCE, FUTURESELFCONT, and COLLECTACTION. In addition to the tertiles, we also show the bottom and upper-5% tails of the estimated HTEs. Each point represents the standardized mean of a variable within a group, and the error bars show the standard error. Variables are standardized (mean = 0, SD = 1) to allow for comparability across scales. Hence, a point at $+1\sigma$ means that the average value of the variable in the specific tertile is one standard deviation higher than the overall population mean. Differences across tertiles are tested using the non-parametric Kruskal-Wallis test [53], with significance thresholds indicated as: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

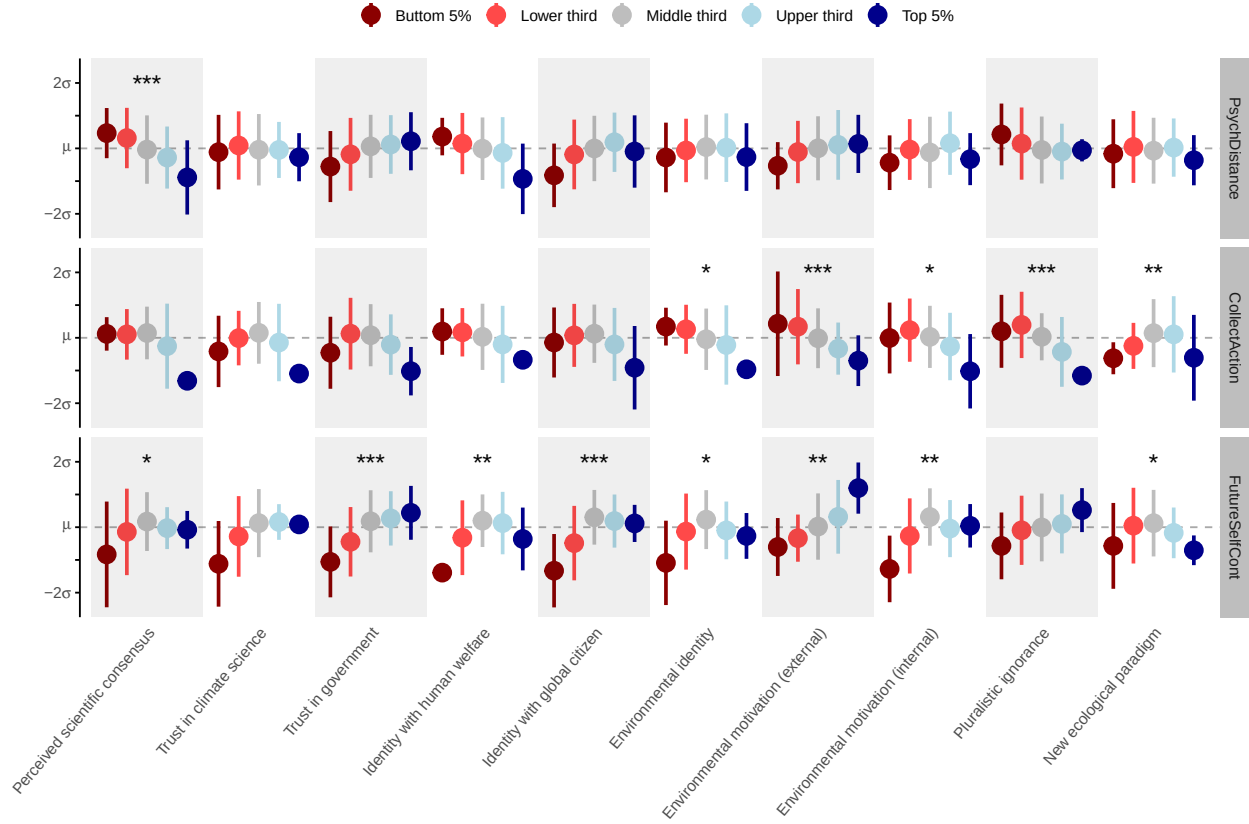


Figure S18: Attitudinal and normative drivers of heterogeneity in the effectiveness of behavioral climate interventions on the outcome climate policy support. Here, we assess whether certain characteristics are systematically associated with higher intervention effectiveness on climate policy support. The plot shows how attitudes and norms are distributed across individuals with different estimated effects of behavioral climate interventions. Here, we stratified the individuals into tertiles by the estimated HTEs for the three interventions: PSYCHDISTANCE, FUTURESELFCONT, and COLLECTIONACTION. In addition to the tertiles, we also show the bottom and upper-5% tails of the estimated HTEs. Each point represents the standardized mean of a variable within a group, and the error bars show the standard error. Variables are standardized (mean = 0, SD = 1) to allow for comparability across scales. Hence, a point at $+1\sigma$ means that the average value of the variable in the specific tertile is one standard deviation higher than the overall population mean. Differences across tertiles are tested using the non-parametric Kruskal-Wallis test [53], with significance thresholds indicated as: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

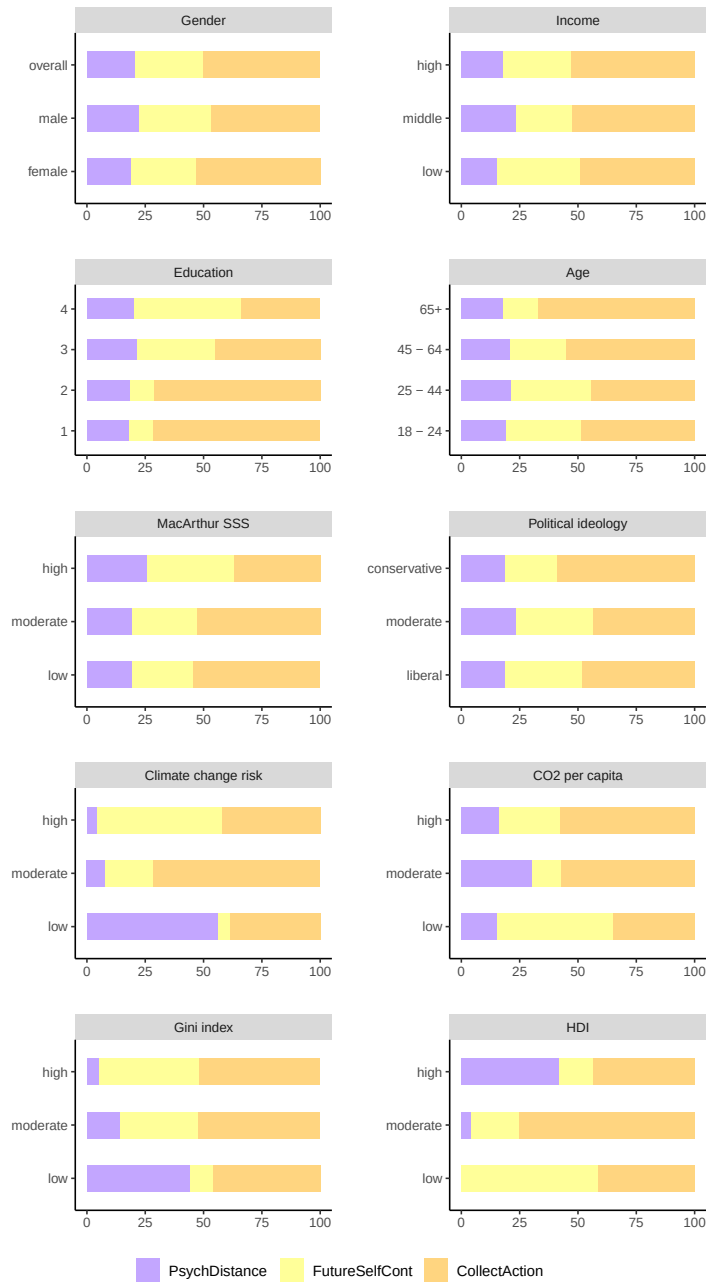


Figure S19: Optimal interventions by sociodemographic subgroups using the outcome climate policy support. The proportions of how often each intervention is optimal (x-axis, from 0 to 100%) and therefore has the largest HTE within that subgroup as determined by the causal forest model. In other words, the optimal intervention is defined as the one with the highest HTE on climate policy support. Education levels are encoded using a discrete categorization (i.e., 1: up to grade school, 2: up to high school, 3: college/undergraduate degree/certificate training, 4: doctorate degree, medical degree, etc.). We categorize the other continuous variables into three equally sized subgroups (e.g., high, middle, and low-level income). We denote the MacArthur scale of subjective social status as MacArthur SSS and the human development index as HDI. For a full description of the sociodemographics, see Supplementary Table S2.

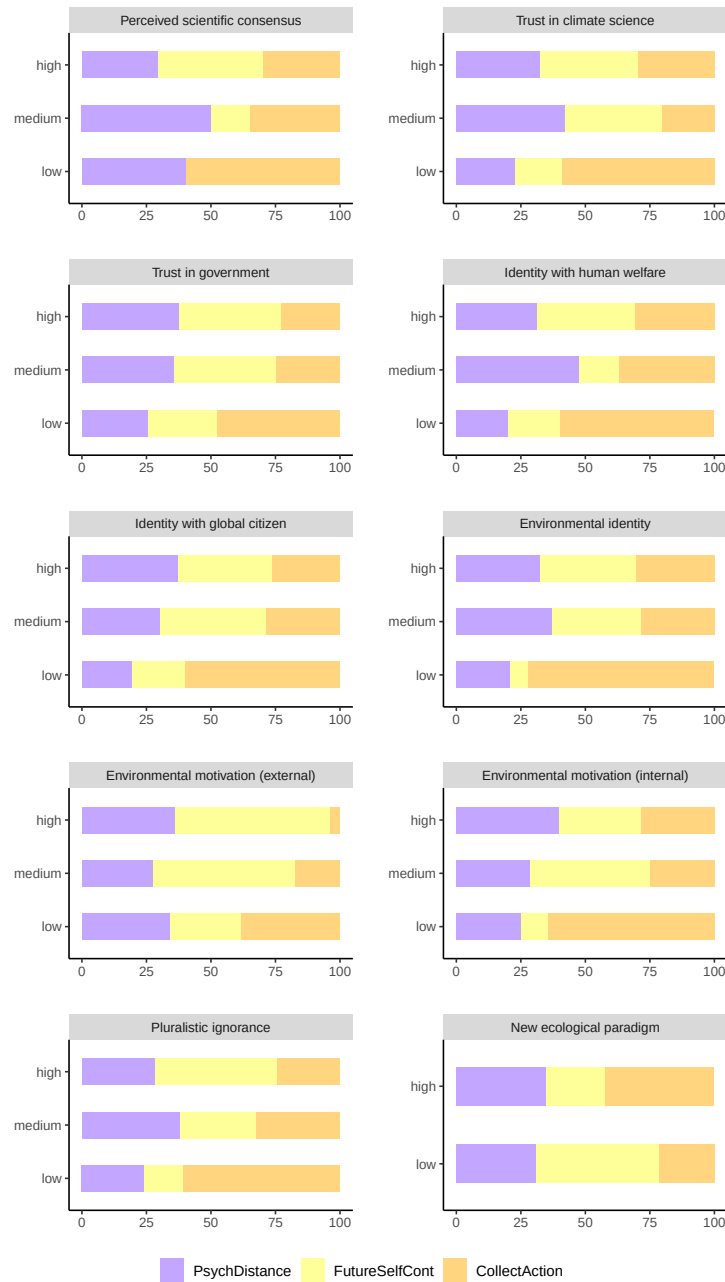


Figure S20: **Optimal interventions by attitudinal and normative subgroups using the outcome climate policy support.** The proportions of how often each intervention is optimal (x-axis, from 0 to 100%) and therefore has the largest HTE within that subgroup as determined by the causal forest model. In other words, the optimal intervention is defined as the one with the highest HTE on climate policy support. We categorize the continuous variables into three equally sized subgroups (e.g., high, middle, and low-level environmental identity). For a full description of the sociodemographics, see Supplementary Table S3.

Supplement B Supplementary tables

Intervention	Description
PSYCHDISTANCE	Frames climate change as a proximal risk using examples of recent natural disasters caused by climate change in each participant's nation and prompts them to write about the climate impacts on their community.
FUTURESELFCONT	Emphasizes the future self-continuity by asking each participant to project themselves into the future and write a letter addressed to themselves in the present, describing the actions they would have wanted to take regarding climate change.
COLLECTACTION	Features examples of successful collective action that have had meaningful effects on climate policies (e.g., protests) or have solved past global issues (e.g., the restoration of the ozone layer).
SYSTEMJUST	Frames climate change as threatening to the way of life of each participant's nation and makes an appeal to climate action as the patriotic response.
LETTER2FUTURE	Emphasizes how one's current actions affect future generations by asking participants to write a letter to a socially close child who will read it in 25 years when they are an adult, describing current actions toward ensuring a habitable planet.
SCICONSENS	Informs participants that "99% of expert climate scientists agree that Earth is warming and climate change is happening, mainly because of human activity."
PLURALIGNORANCE	Presents real public opinion data collected by the United Nations that show what percentage of people in each participant's country agree that climate change is a global emergency.
NEGATIVEEMOTIONS	Exposes participants to ecologically valid scientific facts regarding the impacts of climate change framed in a "doom and gloom" style of messaging that were drawn from different real-world news and media sources.
BINDINGMORAL	Invokes authority (e.g., "from scientists to experts in the military, there is near universal agreement"), purity (e.g., keep our air, water, and land pure), and ingroup-loyalty (e.g., "it is the American solution") moral foundations.
DYNAMICNORM	Informs participants of how country-level norms are changing and "more and more people are becoming concerned about climate change," suggesting that people should take action.
IDENTITYSOCIALNORMS	Combines referencing a social norm ("a majority of people are taking steps to reduce their carbon footprint") with an invitation to "join in" and work together with fellow citizens toward this common goal.

Table S1: Descriptions of the 11 behavioral climate interventions. The behavioral climate interventions were developed by behavioral scientists and pre-screened for cross-cultural feasibility and relevance to climate attitudes [18].

Variable	Description
<i>Individual-level sociodemographic variables</i>	
Gender	<i>What is your gender?</i> ; response options: Male, Female, Prefer not to say, Non-binary/third gender/other.
Income	<i>What is your total yearly family/household income?</i> ; reported using 8 ordinal brackets.
Education	<i>How many years of formal education have you completed?</i> ; categorized into: 0-6 (grade school), 7-12 (high school), 13-16 (college/undergraduate), 17+ (doctorate/professional degrees).
Age	<i>How old are you?</i> ; single-line numerical entry (in years).
MacArthur scale of subjective social status	<i>Please choose the rung where you think you stand at this time in your life relative to other people in your country</i> ; MacArthur Scale from 1 (lowest) to 10 (highest).
Political orientation	<i>What is your political orientation for social issues (e.g., health care, education, etc.) and economic issues (e.g., taxes)?</i> ; each item rated on a 0-100 scale. Items were averaged into a composite conservatism score ($\alpha = 0.84$, see [18]).
<i>Country-level sociodemographic variables</i>	
Climate change risk indicator	Index measuring national exposure to climate-related disasters; sourced from World Bank (https://prosperitydata360.worldbank.org/en/dataset/IMF+RDD). Higher values indicate higher risk.
CO ₂ per capita	Production-based CO ₂ emissions per person (measured in tons); sourced from https://ourworldindata.org/co2-emissions#per-capita-co2-emissions .
GINI coefficient	Inequality index measuring national income distribution; sourced from World Bank (https://data.worldbank.org/indicator/SI.POV.GINI). Higher values indicate greater inequality.
HDI	Human development index combining life expectancy, education, and per capita income; sourced from https://ourworldindata.org/human-development-index .

Table S2: Sociodemographic variables used as predictors of heterogeneity in our analysis.

Variable	Description
Perceived scientific consensus	<i>"To the best of your knowledge, what percentage of climate scientists have concluded that human-caused climate change is happening?"</i> Measured on a scale from 0 to 100.
Trust in climate science	<i>"On average, how competent are climate change research scientists?"</i> ; <i>"How much do you trust scientific research about climate change?"</i> . Measured on a scale from 0 to 100.
Trust in government	<i>"On average, how much do you trust your government?"</i> Measured on a scale from 0 to 100.
Identity with human welfare	<i>"To what degree do you see yourself as someone who cares about human welfare?"</i> . Measured on a scale from 0 to 100.
Identity with global citizens	<i>"To what degree do you think of yourself as a global citizen?"</i> . Measured on a scale from 0 to 100.
Environmental identity	<i>"To what degree do you see yourself as someone who cares about the natural environment?"</i> ; <i>"To what degree are you pleased to be someone who cares about the natural environment?"</i> ; <i>"To what degree do you feel strong ties with others who care about the natural environment?"</i> ; <i>"To what degree do you identify with others who care about the natural environment?"</i> . Measured on a scale from 0 (not at all) to 100 (very much so) and finally averaged.
External environmental motivation	<i>"Because of today's politically correct standards, I try to appear pro-environmental."</i> ; <i>"I try to hide my negative thoughts about pro-environmental behavior in order to avoid negative reactions from others."</i> ; <i>"If I acted anti-environmental, I would be concerned that others would be angry with me."</i> ; <i>"I attempt to appear pro-environmental in order to avoid disapproval from others."</i> ; <i>"I try to act pro-environmental because of pressure from others."</i> . Measured on a scale from 0 (strongly disagree) to 100 (strongly agree) and finally averaged.
Internal environmental motivation	<i>"I attempt to behave pro-environmentally because it is personally important to me."</i> ; <i>"According to my personal values, acting non-environmental is OK."</i> ; <i>"I am personally motivated by my beliefs to be pro-environmental."</i> ; <i>"Because of my personal values, I believe that acting anti-environmental is wrong."</i> ; <i>"Being pro-environmental is important to my self-concept."</i> . Measured on a scale from 0 (strongly disagree) to 100 (strongly agree) and finally averaged.
Pluralistic ignorance	<i>"What percentage of people in your country do you think would agree with the statement: Climate change is a global emergency?"</i> Measured on a scale from 0 to 100.
New Ecological Paradigm (NEP)	Fifteen items assessing general environmental beliefs (e.g., views on human-nature relationships), based on the NEP scale [72]. Widely used to measure environmental attitudes [73]. The items are finally averaged.

Table S3: Attitudinal and normative variables used as predictors of heterogeneity in our analysis

	<i>Dependent variable: Click</i>								
	Overall	Females (18-24)	Females (25-44)	Females (45-64)	Females (65+)	Males (18-24)	Males (25-44)	Males (45-64)	Males (65+)
Personalized	0.273*** (0.020)	−0.044 (0.057)	0.113* (0.057)	0.260*** (0.057)	0.125* (0.055)	0.168** (0.056)	0.127* (0.059)	−0.019 (0.057)	0.820*** (0.058)
Constant	−4.455*** (0.014)	−3.460*** (0.040)	−3.813*** (0.040)	−4.651*** (0.040)	−5.243*** (0.039)	−3.490*** (0.040)	−3.474*** (0.041)	−4.073*** (0.041)	−5.237*** (0.040)
Observations	772,777	43,125	55,033	115,621	237,947	40,559	37,756	74,240	168,496
Log likelihood	−53,538.160	−5,783.487	−5,979.768	−6,810.496	−8,206.704	−5,747.006	−5,327.224	−6,267.137	−7,120.190
AIC	107,080.300	11,570.970	11,963.540	13,624.990	16,417.410	11,498.010	10,658.450	12,538.270	14,244.380

Note: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Table S4: Logistic regression results for the external validation of our personalized targeting strategy on *Google Ads*. Here, we report the overall and subgroup-specific coefficient estimates from our logistic regressions.

Supplement C Related work and theoretical mechanisms

For our main study, we used the three top-performing interventions from [18]. These interventions were: (1) PSYCHDISTANCE, (2) FUTURESELFCONT, and (3) COLLECTACTION. In the following, we provide related work and theoretical explanations for each of these interventions.

The PSYCHDISTANCE intervention [18] frames climate change as a local, immediate, and personally relevant threat. People frequently perceive climate change as psychologically distant—e.g., in space, time, or social relevance—which reduces their urgency of acting [74]. According to *Construal Level Theory*, making climate impacts seem closer, immediate, and more personal increases concrete thinking and, thus, the likelihood of acting [75]. Empirical studies show that localizing and personalizing climate threats significantly increases public concern and willingness to act [76, 77].

The COLLECTACTION intervention [18] focuses on successful climate movements to foster collective efficacy. Collective efficacy—i.e., the belief in a group’s capacity to achieve desired outcomes—strongly predicts engagement in collective actions [78, 79]. Research shows that awareness of past successful movements enhances people’s sense of efficacy and motivates participation in collective climate action [80].

Lastly, the FUTURESELFCONT intervention [18] enhances climate action by increasing the psychological connection to one’s future self. People often neglect distant future outcomes because they perceive their future self as a different person [81, 82]. Strengthening continuity with one’s future self motivates more future-oriented decisions [83]. Empirical evidence confirms that vivid perspective-taking exercises, such as writing letters from one’s future self, significantly increase future-minded behaviors [84]. As such, connecting present behavior directly to personal future consequences fosters responsibility and proactive engagement in climate protection.

Supplement D Background on HTE estimation

Estimand

We adopt the potential outcomes framework [85, 86] to define our causal quantities of interest, namely, the heterogeneous treatment effect (HTE). The potential outcomes framework provides a principled basis for causal inference by comparing the outcomes that an individual would experience under different treatment conditions $t \in \{0, 1, \dots, K\}$, where $t = 0$ denotes the control condition and $K = 11$ corresponds to the different behavioral interventions tested. For each individual i , we define the potential outcome under treatment condition t as $Y_i(t)$. Due to the fundamental problem of causal inference, only one of all potential outcomes is observed for each individual, depending on the assigned treatment condition [85, 87].

Formally, the HTE for intervention t is defined conditional on observed variables X_i , i.e.,

$$\text{HTE}_t(X_i) = \mathbb{E}[Y_i(t) - Y_i(0) \mid X_i], \text{ for } t \in \{1, 2, \dots, K\}. \quad (\text{S1})$$

The HTE thus captures between-subject variability in the intervention effectiveness and thus allows for a more granular understanding of how effects vary across individuals. This is unlike the ATE, which, for intervention t , is defined as

$$\text{ATE}_t = \mathbb{E}[Y_i(t) - Y_i(0)], \text{ for } t \in \{1, 2, \dots, K\}. \quad (\text{S2})$$

Hence, the ATE provides a single “population-wide” estimate of the treatment effect and thus obscures potential heterogeneity in the intervention effectiveness across individuals.

Our analysis uses experimental data in which individuals were randomly assigned to one of the expert-crowdsourced behavioral climate interventions or a control condition. As a result, the assumptions required for valid causal inference—the stable unit treatment value assumption

(SUTVA), positivity, and unconfoundedness—are satisfied by design [85, 86]. This eliminates complex adjustment procedures and ensures identification of the causal estimand and thus adds to the internal validity of the treatment effect estimates.

D.1 Procedure

To estimate HTEs, we applied the causal forest model [38, 39], a nonparametric causal machine learning method designed to capture heterogeneity in treatment effects. Causal forests partition the variable space to capture complex, high-dimensional interactions within the HTEs. In our setup with 11 different treatments, we estimate separate causal forests for each intervention $t \in \{1, 2, \dots, K\}$, each relative to the control group $t = 0$.

To improve the robustness and efficiency of HTE estimation, we apply orthogonalization [88] to our data. Specifically, we leverage the known propensity scores from the random assignment for each $t \in \{0, 1, \dots, K\}$ (i.e., $e_t(X_i) = 0.5$) and estimate the nuisance outcome function $\hat{m}_t(X_i) = \mathbb{E}[Y_i | X_i]$ using a regression forest. This orthogonalization step has several advantages, even in the presence of randomization. First, the climate change belief outcome is non-normally distributed and bounded (e.g., on a Likert or probability scale), making standard parametric assumptions inappropriate [18]. Orthogonalization allows for the use of flexible, nonparametric machine learning methods to model complex relationships in the outcome model. In addition, by partialling out variation from direct contributions of the variables on climate change belief before estimating HTEs, orthogonalization leads to more stable and lower-variance estimates of the HTEs. We then employ the causal forest model to optimize the following objective function based on the orthogonalization:

$$\widehat{\text{HTE}}_t(X) = \arg \min_{\text{HTE}_t(X)} \hat{\mathbb{E}} \left[\left((Y - \hat{m}_t(X_i)) - \text{HTE}_t(X) \cdot (\mathbb{1}_{\{T_i=t\}} - e_t(X_i)) \right)^2 \right]. \quad (\text{S3})$$

D.2 Comparison to ATEs

To estimate ATEs, we use the augmented inverse propensity weighted (AIPW) estimator [45, 46].¹ This nonparametric estimator was chosen for several reasons. First, it offers flexibility in capturing nonlinear relationships and interactions between variables and outcomes. Second, it is robust to violations of parametric assumptions, such as skewness, non-normality, and bounded outcomes, which are present in our underlying data [18]. Third, AIPW is a doubly robust estimator, meaning it yields consistent estimates if either the outcome model or the propensity score model is correctly specified. In our case, the known propensity score from random assignment (i.e., 0.5) guarantees this condition is satisfied. Finally, by incorporating variables through a regression forest in the outcome model, AIPW reduces the variance of ATE estimates compared to simple difference-in-means, even in randomized settings.

The AIPW estimator consists of three components: (i) the propensity score $e(X_i)$ is known here given the randomization in our studies, i.e., $e(X_i) = 0.5$, (ii) the outcome model for the individuals in the control condition is defined as $\hat{m}_0(X_i) = E[Y_i \mid T_i = 0, X_i]$, and (iii) the outcome model of the individuals in one of the interventions $t \in \{1, 2, \dots, K\}$ defined as $\hat{m}_t(X_i) = E[Y_i \mid T_i = t, X_i]$. Using the three components, the ATE for intervention $t \in \{1, 2, \dots, K\}$ can be calculated using the AIPW estimator via

$$\widehat{\text{ATE}}_{\text{AIPW},t} = \frac{1}{n} \sum_{i=1}^n \left[\frac{\mathbb{1}_{\{T_i=t\}} Y_i}{e(X_i)} - \frac{\mathbb{1}_{\{T_i=0\}} Y_i}{1 - e(X_i)} + \left(1 - \frac{\mathbb{1}_{\{T_i=t\}}}{e(X_i)}\right) \hat{m}_t(X_i) - \left(1 - \frac{\mathbb{1}_{\{T_i=0\}}}{1 - e(X_i)}\right) \hat{m}_0(X_i) \right]. \quad (\text{S4})$$

Implementation details

We used the implementation of the causal forest from `grf` (version 2.3.2) [67]. We used 3-fold cross-validation to optimize the hyperparameters of the causal forests. This is done by randomly sampling sets of hyperparameter values, training small forests of size 500 for each, and evaluating their performance using the out-of-bag error. The optimal parameters are then selected by minimizing a smoothed version of this error across sampled configurations. The hyperparameters and

¹We apply the AIPW estimator for both overall ATEs and subgroup ATEs throughout our analyses.

1043 their value spaces are given in Table S5. For all outcome models $\hat{m}_t(X_i)$ with $t \in \{0, 1, \dots, K\}$ we
 1044 used random forests [65] based on the implementation in `grf` [67]. We used the hyperparameters
 1045 in Table S6 for the random forests.

Hyperparameter	Sampled range (from uniform $u \sim \mathcal{U}(0, 1)$)
<i>Minimum node size</i>	$2^{u \cdot (\log_2(n) - 4)}$
<i>Sample fraction</i>	$0.05 + 0.45 \cdot u$
<i>Number of variables</i>	$\lceil u \cdot \min(p, \sqrt{p} + 20) \rceil$
<i>Maximum imbalance of split</i>	$u/4$
<i>Imbalance penalty</i>	$-\log(u)$
<i>Honesty fraction</i>	$0.5 + 0.3 \cdot u$
<i>Honesty prune leaves</i>	TRUE if $u < 0.5$, else FALSE
<i>Number of trees</i>	4000

Note: p refers to the number of variables

Table S5: **Hyperparameters for our causal forests.**

Hyperparameter	Value
<i>Number of trees</i>	2000
<i>Number of variables</i>	$\sqrt{p} + 20$
<i>Minimum node size</i>	5
<i>Maximum imbalance of split</i>	0.05

Note: p refers to the number of variables

Table S6: **Hyperparameters for random forests used for our outcome models.**